Corrected 2 December 2021: See below

science.sciencemag.org/content/373/6551/192/suppl/DC1

Supplementary Materials for

Cauliflower fractal forms arise from perturbations of floral gene networks

Eugenio Azpeitia†, Gabrielle Tichtinsky, Marie Le Masson, Antonio Serrano-Mislata, Jérémy Lucas, Veronica Gregis, Carlos Gimenez, Nathanaël Prunet, Etienne Farcot, Martin M. Kater, Desmond Bradley, Francisco Madueño, Christophe Godin*, Francois Parcy*

*Corresponding author. Email: christophe.godin@inria.fr (C.G.); francois.parcy@cea.fr (F.P.)

Published 9 July 2021, *Science* **373**, 192 (2021)
DOI: 10.1126/science.abg5999

This PDF file includes:

Materials and Methods
Modeling Methods
Figs. S1 to S6
Movie Captions
Tables S1 to S3
References

Other Supplementary Material for this manuscript includes the following:
(available at science.sciencemag.org/content/373/6551/192/suppl/DC1)

Movies S1 to S5 (.mov and .avi)
Archive with Source Code for the Model (.zip)
MDAR Reproducibility Checklist (.pdf)

Correction: Figure S6 and the corresponding explanations in the text have been added to explain (i) how the geometric works, (ii) why some structures were removed, and (iii) the possibility of running a model without removing structures (illustrated by new movies S4 and S5, now added). Revised text appears in red font. In the code archive ZIP folder, the "geometric.model.lpy" file has been replaced with a new file named "geometric.model.erratum.lpy."

MATERIALS AND METHODS

Plant materials

The following *A. thaliana* lines in the Columbia-0 background were used: *lfy-12* (*40*); *ap1-7* (*6*), *cal-1* (*41*), *ap1-7 cal-1* double mutants (gift from J. Goodrich, Institute of Molecular Plant Science, University of Edinburgh, Edinburgh, United Kingdom), SOC1p-GFP (*42*) (provided by D. Posé), *35Sp:FT* (*43*), *35Sp:SOC1* (*44*), *soc1-2* (*44*), *AGL24p:AGL24-RFP* (*45*) and *agl24-2* (*28*). *35Sp:CO* (*46*), *co-3* (*47*), *ft-3* (*43*), *35S:AP1-GR ap1-1 cal-1* (*48*) and *TFL1p:GUS* (this work) *are A. thaliana* lines in the Landsberg *erecta* background, as well as the *35S:AP1-GR ap1-1 cal-1* (*48*) used for the mutation of *CLV3* using the CRISPR-Cas9 strategy.

For crosses between accessions, it was previously verified that the GUS reporter activity was not affected in wild-type Col x L*er* hybrids.

Brassica oleracea var. *botrytis* and Romanesco cauliflowers were obtained from Observatoire Breton des Semences (OBS) or bought at the Grenoble producer market.

Plant growth conditions

Seeds were sown on soil or surface-sterilized and grown in Petri dishes on Murashige and Skoog basal salt mixture medium (Sigma-Aldrich, www.sigmaaldrich.com). Seeds were stratified 1-3 days at 4°C, transferred mixture of phagnum:perlite:vermiculite (2:1:1) or on soil:vermiculite (5:1) and grown at 21-22°C under long-day (16 h) or short-day conditions (8 h). *ap1 cal* plants were grown at 18°C for 16 h light and 16°C for 8 h dark, under 65% humidity. *Nicotiana benthamiana* plants for transient assays were grown on a mixture of sphagnum:vermiculite (1:1) under long-day photoperiod at 24°C.

Plasmids construction

For the reporter analysis of *TFL1* expression, we produced a *TFL1p:GUS* construct that contained the *β-glucuronidase* (*GUS*) gene flanked by 2177 bp of the *TFL1* promoter 5' region and 4605 bp of the *TFL1* promoter 3' region (*20*). The 5' region was amplified from a L*er* genomic clone with primers 5'-TGA<u>GTCGAC</u>GCTAGGAGACTTCGTTGATC-3' (*Sal*I site is underlined) and 5'-CTGCA<u>GGATCC</u>TTTTCTTTTGTTAACTTAGAGG-3' (including the ATG of the *TFL1* gene,

*BamH*I site is underlined). This fragment was cloned as *Sal*I-*BamH*I into the pBI101 vector (*49*), upstream the *GUS* gene. The 3' region was amplified with primers 5'-GTAGAGCTCTAGATTCATGATTGTCATAAACTGC-3' (including the stop codon of *TFL1*, *Sac*I and *Xba*I sites are underlined) and 5'-TAGAATTCGGTACCAAGTTGAAGTCTCTCATTGACGAAC-3' (*Eco*RI and *Kpn*I sites are underlined). This fragment was cloned as *Sac*I-*Eco*RI into the pBI101 vector, thus replacing the nopaline synthase terminator downstream the GUS gene. The resulting *TFL1p:GUS* cassette was cloned as *Sal*I-*Kpn*I into the pBIN19 binary vector. This construct was introduced into wild-type Arabidopsis plants of the L*er* ecotype by vacuum infiltration (*50*, *51*). An homozygous line carrying the transgene in a single locus was selected for further analysis.

For transient expression assays, the cDNAs of *SOC1* and *AGL24* were cloned into the pMDC32 vector by Gateway LR recombination to create *35Sp:SOC1* and *35Sp:AGL24* plasmids respectively. The *35Sp:NGA3* plasmid was already available (*52*). The *TFL1* promoter region IV (fragment between +2823 and + 3230 bp downstream the *TFL1* stop codon) was amplified with forward primer 5'-CTCGAGGACTCTCGAGGACAAACCAAC-3' (*Xho*I site is underlined) and reverse primer 5'-CTTATATAGAGGAAGGGTCTTGATTATGGGTTAGCTATAAAGATGG-3' (overlapping sequence with a 35SminΩ promoter is underlined). The *TFL1* promoter region V (fragment between +3420 and +3752 bp downstream the *TFL1* stop codon) was amplified with forward primer 5´-CTCGAGCGGATTGGTCCAGTTAGAAC-3' (*Xho*I site is underlined) and reverse primer 5'-CTTATATAGAGGAAGGGTCTTGAAGAAGCTCCTACCACTTGAAG-3' (overlapping sequence with a 35Smin (*53*) promoter is underlined). A -90 bp CaMV 35S promoter that includes the omega translational enhancer (35S)(*53*) was amplified with forward primer 5'-CAAGACCCTTCCTCTATATAAG-3' and reverse primer 5'-CCATGGTGTAATTGTAAATAGTAATTGTAATGTTG-3' (*Nco*I site is underlined). The 35SminΩ sequence was fused by PCR downstream the *TFL1* promoter regions with the described region-specific forward primers and the 35SminΩ reverse primer. The resulting fragments were cloned as *Xho*I/*Nco*I into the pGreenII 0800-LUC vector (*54*).

For the CRISPR-Cas9 mediated mutation of *CLV3*, three gRNA spacers specific of the *CLV3* gene were designed using CHOPCHOP https://chopchop.rc.fas.harvard.edu/ and synthetized by eurofins. Guide sequences are the following:

- Guide1: 5'-GATTGGAGACCAGAAGCATCATGA-3'
- Guide2: 5'-GATTGGTTTCTTGGCTGTCTTGGT-3'
- Guide3: 5'-GATTGGTGAATGGGTTGGAGCAAA-3'

These guides were used to build one vector with two gRNA spacers (guides 1-2) and one with the three gRNA (guides 1-2-3)

After annealing oligonucleotide guides, each of the 3 double-stranded DNAs were ligated in pBSK-AtU6-26:guide vector (*55*), modified to allow sequential ligation of several guides (*56*). Resulting plasmids were verified by sequencing. *Kpn*I/*Sbf*I restriction fragments containing the assembled guides and the UBQ10:CoCas9:tUBQ10 cassette of the pBSK:CoCas9 digested by *Sbf*I and *Eco*RI were simultaneously ligated in the pCAMBIA plant transformation vector containing the At2S3:eGFP as selection marker. This plasmid was obtained by cloning the At2S3:eGFP:t35S reporter construct from pFP100 (*57*) into *Bst*X1/*Psp*X1 digested pCAMBIA1300 backbone. The pBSK:CoCas9 contains a codon-optimized version optimized for expression in Arabidopsis of the human Cas9 under the control of the UBQ10 promoter (*56*). Final plasmids (pMLM17 with 2 guides and pMLM18 with 3 guides) were verified by sequencing.

Arabidopsis thaliana **transformation**

Agrobacterium tumefaciens, strain C58C1pMP90, carrying pMLM17 or 18 were grown overnight at 28 °C in Luria Broth medium supplemented with rifampicin 50 µg/mL, gentamycin 50 µg/mL and kanamycin 50 µg/mL and used for the Arabidopsis floral dip method (*58*) using Silwet L-77 at 0.01 %. To improve transformation efficiency of the *35S:AP1-GR ap1-1 cal-1* plants, a spray with 10 mM Dexamethasone supplemented with Silwet L-77 0.01 % was performed 10 days before the agrobacterium floral dip to trigger AP1-GR nuclear translocation and induce flower development and seed recovery.

Molecular characterization of *clv3* CRISPR-Cas9 lines

The effect of the *clv3* mutation in *ap1 cal* were studied by transforming MLM17 and MLM18 constructs in *35S:AP1-GR ap1-1 cal-1*. T1 primary transformed seeds were collected based on their seed fluorescence under a SZX12 Olympus dissecting microscope.

We obtained 14 T1 plants with pMLM17 vector (5 out of 14 showing pyramidal curds) and 16 with the pMLM18 (7 out of 16 showing pyramidal curds). PCR amplification of the *CLV3* genomic sequence around the sgRNA target sites was used to detect Cas9/sgRNA induced mutations and deletions. gDNA was extracted from leaves using Edwards extraction buffer (200 mM Tris-HCl pH7.5 ; 250 mM NaCl, 25 mM EDTA, 0.5 % SDS 20 %). For the PCR amplification, two couples of primers were used oMLM1134/oMLM1131 (166 bp) and oMLM1132/oMLM1133 (268 bp)

oMLM1131: 5'-CATGAGCTTGAGTGAGATCTGG-3'

oMLM1132: 5'-AATGTTGTTCAATTGGCAGATG-3'

oMLM1133: 5'-CCGAAATGGTAAAACCGATAAA-3'

oMLM1134: 5'-GCTACTACTACTACTCTTCTGC-3'

In the T2 generation, non-fluorescent seeds were sown to select against the transgene. The PCR product obtained with oMLM133/1134 primers were sequenced to check mutations at the *CLV3* locus. Sequencing revealed several types of mutations in the *CLV3* gene. Plant MLM18-25-4 has a T nucleotide insertion at position 57 creating a frame-shift with a stop codon in the 1st exon ((First exon sequence with added t and bold stop codon: ATGGATTCGAAGAGTTTTCTGCTACTACTACTACTCTTCTGCTTCTTGTTCCTTCAt**TGA**) amino acid sequence: MDSKSFLLLLLLFCFLFL**H**), as well as a deletion of 100 bp between guides 3 and 2. The mutant MLM17-3 shows the same insertion of a T nucleotide as the MLM18-25-4 mutant. Plants shown in figure 4 are MLM18-25-4 T3 (Fig. 4j), MLM17-3 T1 (Fig. 4k) and MLM18-4-28 T3 (Fig. 4m).

To verify that the pyramidal phenotype observed in *ap1 cal* 35S:AP1-GR is due to a single mutation, one T2 plant (MLM18-2-19 in *35S:AP1-GR ap1-1 cal-1*) was crossed to *35S:AP1-GR ap1-1 cal-1* and F2 progenies from two individual F1 plants were analyzed. The pyramidal phenotype was observed in 6 out of 35 (17%) and 8 out of 40 (20%) F2 plants consistent with a 25% / 75% segregation with a single mutation at the *CLV3* locus (Chi-square test of independence $p = 0.205 > 0.5$).

As a control, pMLM18 construct was transformed into the Col-0 and L*er* wild type backgrounds. 27 and 30 T1 plants were obtained in Col-0 and L*er* backgrounds respectively, all showing typical *clv* phenotype including increased petal number, increased pistil size due to increased carpel number and inflorescence meristem size.

Seed and plant imaging

For selection of MLM17 and MLM18 transgenics based on At2S3:GFP fluorescence in seeds, an Olympus SZX12 dissecting microscope with fluorescent illumination was used. Light micrographs were taken with the Keyence digital microscope (VHX-5000) using a Z20 X50 lens.

Confocal imaging was performed as described (*59*). Environmental scanning electron microscopy experiments were performed at the Electron Microscopy facility of the ICMG Nanobio-Chemistry Platform (Grenoble, France). Untreated flowers were directly placed in the microscope chamber. Secondary electron images were recorded with a Quanta FEG 250 (LV) microscope while maintaining the tissue at 2 °C, under a pressure of 120 Pa and a 25 % relative humidity. The accelerating voltage was 7 kV and the image magnification ranged from 100 to 800X.

GUS assay and histological procedures

Arabidopsis shoot apices were stained for GUS as described (*20*) with minor modifications. Staining buffer contained 10 mM of potassium ferri- and ferrocyanide in order to minimize leakage of the X-Gluc reaction product (*60*). Photographs of whole-mount samples were taken with a Color View 12 digital camera connected to a Nikon SMZ800 binocular stereoscopic microscope. After GUS staining, some samples were cleared with chloral hydrate solution for 2-3 days at 4 °C (*61*). Images of cleared apices were taken with a Nikon DS-Fi1 digital camera connected to a Nikon Eclipse 600 light microscope under Nomarski interference optics. For tissue sectioning, samples were stained in 0.2% (w/v) eosin and embedded in paraffin as described (*20*). 12 µm sections were obtained with a Leica RM-2025 microtome and images were taken under bright-field microscopy.

Luciferase transient assays

4-week-old Nicotiana plants (with 5-6 leaves) were used for the analysis. Agrobacterium cultures (C58pMP90 strain) were incubated for 24 hours at 28 °C (to stationary phase, OD_{600} 1-2). Cells were collected and resuspended in infiltration buffer (10 mM $MgCl_2$, 10 mM MES and 0.2 mM acetosyringone) to an OD_{600} of 0.4. Agrobacterium mixtures with a 1:5 reporter:effector ratio were prepared. These were incubated for 3 hours at room temperature and darkness with gentle shaking (50 rpm) before infiltration of young and fully expanded Nicotiana leaves. Three days later, firefly luciferase (LUC) and renilla (REN) activities were assayed from 0.5 cm leaf discs (approximately 20 mg) using the Dual-Luciferase® Reporter Assay System kit according to the manufacturer's

instructions (Promega). Absolute relative luminiscence units were measured with a GloMax™ 96 Microplate Luminometer (Promega). LUC/REN ratios were averaged from at least three biological replicates (different plants) each one with three technical replicates (different leaves from the same plant).

Chromatin immunoprecipitation assay

The commercial antibody Living Colors_DsRed polyclonal antibody (Clontech) and an antibody against a synthetic peptide (SVKCIRARKTQVFK) were used for AGL24 and SOC1 ChIP experiments respectively. Chromatin was prepared from inflorescences and three independent biological replicates were done. Wild-type (Col-0) plants were used for chromatin extraction for the SOC1 ChIP experiment, and *soc1-2* plants served as a negative control. Homozygous *AGL24p:AGL24-RFP agl24-2* plants were used for the AGL24 ChIP experiment and wild-type plants served as a negative control. The ChIP assays were performed as described previously (*62*). Six primer sets (fragments 1-6) were designed in the 3' *TFL1* promoter region: the forward primer 5'-AAACGTGGAGATACGGAAAAGG-3' and the reverse primer 5'-ACCAGCCGTGAAAATAGATATG-3' for fragment 1, the forward primer 5'-GCATTCTACATTGATTCAGTG-3' and the reverse primer 5'-TGAATTAATGACACGTGACC-3' for fragment 2, the forward primer 5'-GTTTTAGGGTTTCAGTAACAC-3' and the reverse primer 5'-ATGGAATGGAACAGAGCACG-3' for fragment 3, the forward primer 5'-GGTCCAAGGGTTAGTATGTTTC-3' and the reverse primer 5'-GCCGCAAACCTGGTGATTAACC-3' for fragment 4, the forward primer 5'-GAAACATACTAACCCTTGGACC-3' and the reverse primer 5'-GTAAATGTACCTCCTCGTCAC-3' for fragment 5, and the forward primer 5'-TCAATTTCGGATTGGTCCAG-3' and the reverse primer 5'-CTTAGTTGTAACTGATGAACG-3' for fragment 6. Enrichment folds were detected by qPCR using a SYBR Green assay (Bio-Rad). The assay was performed in triplicate using a Bio-Rad C1000 Thermal Cycler optical system and relative enrichments, with their associated relative errors, were calculated as previously described (*62*).

Cauliflower curd mRNA sequencing

Curds from developing cauliflowers grown in the field (planted July 10th 2017, harvested Oct 18-27 2017) were provided by O.B.S (Breton Seed Observatory): OBS_5045_1 (named OBS-Caul-1 on figure S5), OBS_800_1 (OBS-Caul-2), OBS-0795_2 (OBS-Caul-3), OBS_800_2 (OBS-Caul-4), OBS_0795_1 (OBS-Caul-5), OBS_0819_1 (OBS-Caul-5). We also used two cauliflowers (Samples CAUL_1, CAUL_2a, CAUL_2b) and two Romanesco (Samples ROM_1, ROM_2a, ROM_2b) bought on the Grenoble producer market in 2017. 2a and 2b correspond to extractions made on different florets from the same curd. Total RNAs were extracted using the RNeasy Plant Kit protocol (QIAGEN) followed by DNAse treatment and sent to Genewiz (UK) for quality control, Poly-A selection, library cloning and sequencing (Illumina HiSeq4000, 2x150bp configuration, single index).

The obtained reads were aligned with the HDEM Reference Genome (*63*) with STAR (outFilterMultimapNmax 1; outFilterMismatchNmax 6; alignIntronMax 30000; other parameters set by default). FeatureCount (default settings) was used to detect reads mapping to *Brassica oleracea* genes. EdgeR (default settings) was then used to identify the genes differentially expressed between the individuals analyzed. The orthologs of *AP1* (At1g69120), *CAL* (At1g26310) and *FUL (*At5g60910) of cabbage are identified by BLAST protein against the *Arabidopsis thaliana* proteome (TAIR10). The Heatmap of the expression of these genes is generated by a python ad hoc script. Comparisons of the genes expressions on multiple tissues were done using datasets from GSE42891 (*64*). Orthology between our reference genome (HDEM) and their reference were done with reciprocal Blast and selection of the best candidates.

The following secure token has been created to allow review of record GSE150627 while it remains in private status: khkjgckmdtkhpgb.

***BoAP1* promoter analysis**

BoAP1-a/c and *BobAP1-a/c* promoters were extracted from HDEM (*63*) and D-134 (*65*) reference genomes, respectively. The sequences were scanned with the Position Weight Matrices (PWM) of three transcription factors known to regulate *AP1* in Arabidopsis. PWM were taken from the JASPAR database for FD and SOC1 (*66*) or from (*67*) for LFY. Based on previous analyses on LFY (*67*), we chose a score threshold of -18 to distinguish relevant TFBS. We calculated

thresholds for FD (-8.5) and SOC1 (-9.5) to achieve similar TFBS frequency as for LFY sites with score > -18. The PWM and computed thresholds were used to identify TFBS in *BoAP1-a/c* and *BobAP1-a/c* promoters.

MODELING METHODS

Regulatory network construction and simulation

SALT network

The SALT model included the *SAX* genes (i.e., *SUPPRESSOR OF OVEREXPRESSION OF CONSTANS 1* (*SOC1*), *AGAMOUS-LIKE 24* (*AGL24*) and *XAANTAL2* (*XAL2*)), *APETALA1* (*AP1*) and *CAULIFLOWER* (*CAL*) (hereinafter *AP1/CAL*), *LEAFY* (*LFY*), and *TERMINAL FLOWER1* (*TFL1*). These genes are the key transcription factors (TF) or co-factors that have been experimentally associated with inflorescence and flower development and the development of cauliflower structures in *A. thaliana*. We also included two inputs, auxin, a phytohormone that promotes the initiation of new primordia, and F that represents different types of molecules triggering flower production. Because we did not include the auxin signaling pathway and F possibly represents multiple flowering inducing pathways, the activity of both auxin and F is only phenomenological in our model. Auxin concentration is maximal in newly formed primordia, while its activity is basal in the shoot apical meristem. For this reason, shoot apical meristems and newly formed primordia states are different in their auxin levels. As the value of F increases during the plant development in every meristem, we expect the shoot apical meristem to transit to an inflorescence state, and to a flower state for lateral primordia. The network includes multiple positive and negative regulatory interactions listed in Table S1. We model all regulatory interaction using Hill equation of the form:

$$\frac{dY}{dt} = \frac{X^n}{k_{X,Y}^n + X^n},$$

for positive regulations and:

$$\frac{dY}{dt} = \frac{k_{X,Y}^n}{k_{X,Y}^n + X^n},$$

for negative regulations. $k_{X,Y}^{n}$ is the dissociation constant or binding affinity of protein *X* to the promoter of gene *Y*, which is equal to the concentration of free TF at which half of all binding sites are occupied and n is the Hill coefficient that usually represents the number of TF molecules that are required to bind to a gene regulatory region, also known as level of cooperativity. Using Hill equations to represent the regulatory interactions we build the GRN described by the next system of equations:

$$\frac{dS}{dt} = \left(\frac{k_{A,S}^{n}}{k_{A,S}^{n} + A^{n}}\right)\left(\frac{F}{k_{F,S} + F} + \frac{S^{n}}{k_{S,S}^{n} + S^{n}}\right) - \left(k_{a_{S+A}}SA - k_{d_D}\right) - \delta S$$

$$\frac{dA}{dt} = \left(\frac{k_{T,A}^{n}}{k_{T,A}^{n} + T^{n}}\right)\left(\frac{L^{n}}{k_{L,A}^{n} + L^{n}} + \frac{S^{n}}{k_{S,A}^{n} + S^{n}}\frac{F}{k_{F,A} + F}\right) - \left(k_{a_{S+A}}SA - k_{d_D}\right) - \delta A$$

$$\frac{dL}{dt} = \left(\frac{k_{T,L}^{n}}{k_{T,L}^{n} + T^{n}}\right)\left(\frac{aux + aux_b}{k_{aux,L} + aux + aux_b}\frac{k_{L,L}^{n}}{k_{L,L}^{n} + L^{n}} + \frac{A^{n}}{k_{A,L}^{n} + A^{n}}\right) - \delta L$$

$$\frac{dT}{dt} = \left(\frac{S^{n}}{k_{S,T}^{n} + S^{n}}\frac{k_{R,T}^{n}}{k_{R,T}^{n} + R^{n}}\frac{k_{D,T}^{n}}{k_{D,T}^{n} + D^{n}}\right) + \frac{L^{n}}{k_{L,T}^{n} + L^{n}} - \delta T$$

$$\frac{dD}{dt} = k_{a_{S+A}}SA - k_{d_D} - \delta D$$

$$\frac{daux}{dt} = -\delta aux$$

$$\frac{dR}{dt} = -\delta R$$

S, *A*, *L*, *T* and *aux* represent *SAX*, *AP1*, *LFY*, *TFL1*, and auxin respectively. *D* and *R* represent the dimer *SAX-AP1*, and the eREP repressor of *TFL1*. aux_b represents a "basal level of auxin activity". For sake of simplicity, aux_b includes the synthesis, import and signaling in an apical meristem and is considered constant. δ is the degradation rate. Here, in $k_{X,Y}$, $X \in \{S, A, L, T, D, R, F, aux\}$ and $Y \in \{S, A, L, T\}$. $k_{a_{S+A}}$ and k_{d_D} are the association rate of AP1 with SAX and the dissociation rate of the SAX-AP1 complex. We optimized the SALT model choosing different values of δ, $k_{X,Y}$ and *n* within sensitive biological values (see Parameters below).

Parameters

Data for the exact value of the parameters included in the network are not available in the literature. However, we used general knowledge about the different processes to constrain the parameter space. In particular, we assumed that the degradation δ of all variables were equal to *0.1*/minute, which is a realistic value given that the half-life of proteins is usually within the hours range (*68*). The level of cooperativity n for all X and Y genes were equal to 2, representing the formation of

protein dimers for gene regulations, which is common for gene regulation (*69*). Finally, we considered that the concentration of TFs in cells is within the nanoMolar (nM) range (*70*). The affinity of transcription factor to the DNA is usually lower than 10^{-8}M (i.e., affinity≤μM)(*71*). Finally, TFs or cofactors, including TFL1, AP1, and LFY bind to hundreds of target genes, and all TFs can randomly bind to the DNA (32). Thus, the effective concentration of TFs is lower than the real concentration. Assuming that the effective concentrations of the TFs is 2 orders of magnitude smaller than the real concentration, we constrain $k_{X,Y}$ values within the range $[10^{-10}\text{M}, 10^{-6}\text{M}]$.

The SALT model was able to produce the expected behavior, and the kinetic was optimized by hand (see Table S2) leading to "optimal" values of the parameters. Then, we analyzed the effect of modifying the value of the $k_{X,Y}$s. To do this, we fixed the values of all $k_{X,Y}$ to their optimal values, except one, that we varied within the interval $[10^{-10}\text{M}, 10^{-7}\text{M}]$. As observed in Fig. S2, the overall behavior of the network was maintained for most $k_{X,Y}$ values, showing that the dynamic of the network is robust to changes in the values of the parameters.

Structure of the model

The model structure (as reported in Fig. 3a, with feedback loops and feedforward loops (FFL) listed in Table S3), allows for some analysis which is independent of specific parameter values. The "mutual repression" positive feedback listed in Table S3 is solely mediated by AP1. The other positive 2-loop, between AP1 and LFY, is visible but cannot affect the competition with TFL1: it plays a role of stabilization, or self-maintenance, of the floral identity. One consequence of this loop is that, at steady state, LFY and AP1 are always expected to be either both high or low. It is apparent, without having to write any detailed model of the dynamics, that the removal of AP1 above deprives the network from its ability to overcome TFL1's expression; even if external signals transiently increase LFY's expression, without AP1's contribution this cannot be maintained permanently and the system must reverse to a ``non-flower" identity (high TFL1). The structure of the positive three loop entails that any steady state must not only have AP1 and LFY at similar levels, but that this should be opposite to TFL1 level, again confirming the antagonism floral/non-floral identities.

Coherent FFLs have been characterized (*72*) to induce a response delay (to "off steps" for type 2 and "on steps" for type 4, given the AND nature of these FFLs), which can have the function of a

"persistence detector" (i.e. their last term responds to pulses of the first term only if it persists over time), while incoherent FFLs have the potential to speed up response times (the "on step" for type 1 and "off step" for type 3, with also a potential to induce pulses of TFL1 if LFY is present). For the SALT model this leads to the following interpretation: the GRN comprises a core "ALT" network, which has an intrinsic ability to present clearly distinct floral (high LFY and AP1, low TFL1) and non-floral attractors (high TFL1, low AP1 and LFY), while also presenting homeostatic properties able to transiently maintain the network in intermediary states (all three core variables at intermediate levels). On top of this core network, a layer of feedforward loops is able to speed up the triggering of the core ALT network's internal dynamics, while protecting the flower inducing genes from transient removal of the florigen signal. Without the AP1 node, the remaining LFY-TFL1 "core" network is irremediably deprived of its bistability, and only high TFL1/low LFY can be achieved as a permanent state.

Two remaining FFLs (6th and 11th in Table S3) have the potential to make the increase (in response to SAX, i.e. F after a delay) of LFY faster than that of TFL1 and make the removal of TFL1 (upon SAX removal) delayed compared to that of LFY. Both can contribute to the occurrence of a transient pulse of LFY upon F induction via SAX, during which TFL1 remains low.

Introducing a factor X

Even though the SALT network has the ability to extend the persistence of a transient pulse of LFY in time, this can only be a reversible process; the irreversible effects produced by the LFY pulse (i.e. the development of branches of order >2 and the removal of the bract) observed in the *ap1 cal* mutant cannot be explained with the SALT variables alone. The most parsimonious explanation requires some transcriptional target of LFY, say a factor "X" (which could actually correspond not only to one but to several transcriptional targets, aggregated into a unique factor X here), which would be permanently activated by a transient exposure to LFY. This requires this factor X to either self-activate, or to be part of a larger positive feedback loop involving additional transcription factors. Choosing the former for parsimony, one would typically have

$$\frac{dX}{dt} = \left(\frac{L^n}{L^n + k_{XL}^n} + \frac{X^n}{X^n + k_{XX}^n}\right) - \delta_X X$$

Given the hypothetical nature of this equation, we simplify it to a limit where the Hill function becomes a hard step (i.e. $n \to \infty$), so the production rate (term in brackets) is a Boolean like "1 if either X or L is high, 2 if both are high". The finite *n* case would be qualitatively equivalent, but requires more parameter values and does not lead to closed form solutions. In this model, X has two stable values, high and low, and the high value can be triggered permanently if LFY remains high for some time, i.e. $L(t) > k_{XL}$ for a long enough interval of time values (say at least Δt, defined as the time for X to reach its threshold k_{XX} from 0).

In practice, this is strictly equivalent to a simpler criterion, where LFY is compared to a threshold value larger than k_{XL} calculated to ensure it takes at least Δt time units for LFY to decrease down to k_{XL}. We have therefore chosen this simple implementation. In summary: the factor X is a Boolean variable, which is set to 0 initially and switches irreversibly to 1 if LFY reaches a threshold value at one point during its history. This was incorporated in the model with the piecewise function:

$$\begin{array}{lll} X(t) = 0 & if \;\; X(t - dt) = 0 \quad \& \quad L < L_{Xth} \\ X(t) = \;1 & otherwise \end{array}$$

where L_{Xth} represents the LFY threshold needed to activate X (Table S2). The equations describing the SALT model plus this one are the equations used in the plant architecture and GRN model described in the next section.

Coupling of plant architecture model and GRN

To model the interaction between the GRN studied in this work and the development of plant architecture, we built a model for organ and branching system development and coupled this developmental model with the GRN model presented in the previous section.

Our developmental model consists of a set of developmental rules expressing the growth of the plant apices and the growth dynamics of the different organs. These rules are encoded using the L-system formalism (*73*) in the L-Py programming language (*74*). The model is a context-free deterministic L-system (D0L-system), where components change according to physiological and genetic rules depending on their own state, but independent of the state of their neighbors in the plant.

Meristem state. We assume in our model that plant apices i (i.e. meristems) have a state s_i composed of molecular variables (levels of gene expression, hormones, etc.), geometric variables (radius, order), and physiological variables such as age, time count since last production, dormancy, physiological age, number of plastochrons already produced, duration in the current physiological state.

At a macroscopic level, meristems can be in one of three categories of physiological state corresponding to three broad classes of the detailed states s_i: VEG for vegetative, INFLO for inflorescence, FLO for flower. The first meristem starts in a VEG physiological state. During growth, new meristems are initialized with respect to the parent meristem state s_i: they inherit all the values LFY, AP1, SAX of their parent meristem, a value of TFL1 multiplied by a fraction factor (by default 0.1) and maximal level of auxin and of TFL1 repressor (eRep). The value of F is also inherited with potentially an attenuating factor (by default 1 - no attenuation).

Rules determining meristem macroscopic physiological state. At any time *t*, meristems that have a state with high LFY and high AP1 are set to FLO physiological state. Meristems that are not in FLO physiological state can either be in VEG or INFLO physiological states. A VEG physiological state corresponds to a low sum of SAX and LFY, while INFLO physiological state corresponds to a high value of the sum of SAX and LFY. The fact that the sum of LFY and SAX determine the physiological state reflects the fact that LFY and SAX induce bolting independently. Low/high values of specific gene activities are determined with respect to specific thresholds parameters. These thresholds were set so that in WT plants they lead to realistic numbers of organs and development (inspired from Mündermann et al. (*75*)).

Meristem growth. In VEG physiological state, meristems produce a short internode, an axillary leaf at a phyllotaxis angle of 137 degrees from the preceding one at each plastochron. Axillary meristems are forced to stay dormant (we do not model the outgrowth and dynamics of lateral meristems in the rosette). In INFLO physiological state, a bract/cauline leaf is produced if the X factor (reflecting a memory of high levels of LEAFY) is not set in the current meristem state, together with an axillary meristem. Finally, as soon as a meristem enters a FLO state, it immediately transforms into a terminal flower with age 0.

Conditions for lateral productions to grow out: The meristem building the main stem (order 0 in the code) grows immediately after initiation. Lateral meristems in the rosette meristem never

grow (their growth is not modelled). Meristems building stems of order 1 (lateral branches) always start to grow with a small and constant delay with respect to the moment of their initiation (half a day by default). Other meristems (at higher orders) are latent by default (in reference to what is observed in WT, presumably due to apical dominance exerted to apices of order 1 on meristems of higher orders). This dormancy is cancelled if meristems have experienced a high level of LFY since their initiation, which by triggering the factor X (see above), induces meristems outgrowth (case of cauliflower).

Organ growth. Every generated organ grows during the simulation according to its own growth rules. These rules depend on the physiological state of the meristem that produced that organ. For instance, an internode produced by a meristem in VEG physiological state will not elongate, as opposed to meristems produced by meristems in INFLO or FLO physiological states. Growth rules are essential to the dynamics of architecture development, in particular the rules controlling internode elongation. These rules have been implemented to reproduce qualitatively realistic rosette dynamics, bolting and flowering development on the WT. We assume here that these rules are not modified by the mutations affecting the inflorescence studied here. The cauliflower architecture thus emerges only from a change in the genetic network and its consequence on organ outgrowth or suppression, with a conserved dynamics of axes elongation.

Gene regulation computation. At each time step, each meristem reads the new value of F, whose dynamics is considered as a global input variable to the plant and determined here by a growing sigmoidal function of the current development time. This value is used as an input of the genetic network, together with the previous state of the meristem to update the genetic state of the meristem. This new genetic state may in turn determine a new value of the macroscopic physiological state of the meristem and induce a change in meristem behavior according to previously defined rules.

A consequence of the meristem development rules and the monotonic increase of F, all meristems follow a logical sequence of differentiation by moving to the next physiological state after a while:

$$\text{VEG} \rightarrow \text{INFLO} \rightarrow \text{FLO},$$

thus producing a progressive macroscopic differentiation of the meristems. In the cauliflower, this differentiation is blocked in the INFLO state as LFY can never set up firmly the flower state

L-systems implementation principle. At any moment during a simulation, the simulated plant is represented as a set of components organized in a bracketed string of modules and called the *L-string*. Modules represent the various plant components (meristems, internode, leaves, etc.) that can be found in a plant branching system. Each module can bear attributes describing its state (e.g. diameter, length, genetic state, etc.). In the L-string, modules within a pair of matching brackets correspond to the successive components of a given axis. The position of an opening bracket in the L-string defines the position of the axis in the tree branching system. Altogether, the L-string represents the state of the plant branching system at each time step.

For example, consider a L-string made of three main types of components denoted *I* (internodes), *A* (apices), *L* (leaves):

L = $I(5,21)I(5,19)[L(10,40)][I(3,10)A(s_1)]I(4,12)[A(s_2)]A(S_0)$

It represents a plant made of 3 axes:

1. $I(5,21)I(5,19)I(4,12)A(s_0)$
2. $I(3,10)A(s_1)$
3. $A(s_2)$

Leaves, such as $[L(10,40)]$, are considered as insertions on their own axis. In this example, internodes *I* have two attributes, namely *d* and *l*, corresponding respectively to their average diameter and length. Likewise, leaves *L* have two attributes corresponding to their maximum width and length. Apices *A* have a single vector attribute aggregating several variables defining the state of an apex. *A* typical such state contains variables corresponding to plastochron information, gene expression and signal levels.

Let's assume for instance that the state of each apex is governed by a GRN composed of 3 genes, A, B, C. Then the state of every apex will contain a value corresponding to the level of expression of these genes in the apex. In our example, each state s_i of apex i will contain the following variables: $s_i.A$, $s_i.B$, $s_i.C$, corresponding to the expression levels of genes A, B, C in state s_i, and a plastochron count down value $s_i.PCD$, counting the time remaining before initiating a new lateral organ and reset at the initiation of each lateral primordium.

Based on these definitions, assuming a constant increment of time Δt between the simulation steps, the rules of development have the following main structure:

$A(s)$: *# apex production rule (s = state)*

compute the new apex state ns as a function of the old state s and Δt

(this updates in particular the apex GRN variables, including factorX)

if $s.PCD == 0$: *# time to make a new primordium*

compute an initial state is for the new lateral apex

reset $s.PCD$ to T *# i.e. to a full plastochron*

if factorX < fXthreshold**: produce** $[A(is)]A(ns)$

else: **produce** $A(ns)$ *# the bract is suppressed*

else:

decrease $s.PCD$ by Δt

produce $A(ns)$

$I(y, l)$: *# internode production rule, (y = diameter, l = length)*

compute dy and dl as a function of the increment of time Δt

produce $I(d + dy, l + dl)$

$L(w, l)$: *# Leaf production rule (w = width, l = length)*

compute dw and dl as a function of the increment of time Δt

produce $L(w + dw, l + dl)$

The first rule defines the time evolution of an apex. First the new state ns is computed based on the apex previous state s and on Δt. This updates the value of the gene expression levels contained in the apex current state (detailed below). Then, if the plastochron countdown has reached 0 ($s.PCD == 0$), it triggers the production of a new lateral apex with an initial state partially inherited from the main apex's state (see above), and an apical state with updated state. Note that depending on the value returned by the GRN for the factor X, the network will keep or suppress the bract (this illustrates how gene levels are used to take decisions at the level of morphodynamics rules). Otherwise, one just replaces the current apex with an updated one.

The last two rules define how each internode I and leaf L grow with time. Increment of the geometric variables (length, width and diameter) characterizing these components are computed over a time lapse Δt, and are used to update the values of internode and leaf geometric variables.

Finally, a set of interpretation rules make it possible to map at each time point the current L-string to a detailed geometric interpretation. The detailed description of these rules can be found in the available code.

Coupling with the GRN model

At time step t, each apex i is in a state s_i stored as an attribute of the corresponding component $A(s_i)$ in the L-string. We assume that this state is characterized by some attractor of the SALT GRN described in previous section *Regulatory network construction and simulation*. During development, this state can change if one of its entries, aux or F, is forced to a new value by the architectural processes implemented in the L-systems rules and external to the GRN *per se*.

Let us denote $g_1, \dots, g_k$ the set of GRN variables and call $g_1, \dots, g_{k_0}$ subset of the entries that might be overwritten by external processes during the plant growth (variables that can be forced from the outside in the GRN). At the end of the nth time step, the state of an apex i is stable, defined by s_i, and the kth gene having the value $s_i.g_k$. During the next step of duration Δt, apex processes or aging may change the value of one or several of the input gene variables $g_1, \dots, g_{k_0}$. In this case, the apex GRN state from the previous state may no longer be valid, i.e. compatible with the GRN model. Therefore, the GRN transition function must be called with the new GRN input gene values to compute the attractor corresponding to this new situation. All these updates are done at the beginning of the Apex production rule (see above).

Between two simulated steps, the state of every apex is thus evaluated using this rule. The state of newly initiated lateral meristems is derived from the apical meristem state, using a set of putative inheritance rules (see description above). Thanks to this tight coupling between the plant architecture model and GRN, the architecture of the plant can be seen as an emerging property of the two-way interaction between GRN and growth. It made it possible to reproduce all the mutant phenotypes related to genes involved in the cauliflower GRN.

Geometric model

In addition to the GRN-based plant development model, we built a purely architectural or geometric model (no GRN involved) to study how changes in the plastochron modify the morphology of curds. This model is a simplified version of the plant architecture model described above.

In this model, a meristem (represented by a sphere of constant size) produces an elongating internode and a lateral meristem at each plastochron. A newly born meristem starts producing its own lateral meristems after a time defined by the parameter called Onset Delay (OD). Increasing OD values decreases the total number of meristems produced by the simulation. OD = 0 was used in Figure 4F-G. Internodes elongate in time according to a common sigmoidal curve. Starting from an initial meristem, the repetitive application of these rules generates a curd composed of branching systems called florets (fig. S6A). To avoid the high computational cost after some time due to the production of an exponentially increasing number of elements, we simulated meristems and internodes up to a chosen maximal order (MO). Florets of order = MO were represented by an axis bearing a sphere used as envelope to coarsely approximate the non-simulated internodes and meristems of higher order (fig. S6A). However, for long simulation times, the growth curve of the envelopes creates large structures that become poor approximations of the oldest florets and alter the actual curd structure (fig. S6A and orange spheres in fig. S6B). For a given number of simulation days, more precise curd structures are obtained by increasing the detail level (i.e. by increasing the MO value): fig. S6B shows that the envelopes become a better approximation as they are used for younger florets when MO is increased. To limit the visual impact of poorly simulated floret envelopes at low MO, we thus designed an empirical rule that removes those large elements artificially altering the curd shape. This produces a curd structure closer to the fully detailed one (i.e. with all orders simulated, Fig S6C). For example, removing the oldest envelopes from the MO=4 model makes it closer to the finer one obtained at MO=9 (fig. S6B).

Limiting simulation time or increasing the OD parameter make it possible to reduce the number of simulated meristems. In such cases, all meristems can be simulated at all orders and one can optionally choose to inactivate the previous empirical rule and represent all meristems by spheres of constant size (fig. S6D-E).

These two options of the geometric model show that cauliflower and Romanesco morphologies can be produced by changes in the plastochron duration (Fig.4F-G, fig. S6D-E). To produce a cauliflower morphology, the plastochron remains constant during the entire simulation (10 h). For the Romanesco morphology new meristems are initialized with a 10 h plastochron. Then, the duration of the plastochron linearly decreases as the meristem grows.

The model was implemented as an L-system (see previous section). For the cauliflower morphology the plastochron countdown decreased following:

$$s.PCD(t + \Delta t) = s.PCD(t) - cte$$

while for the Romanesco the plastochron countdown decreased faster with the age of the meristem:

$$s.PCD(t + \Delta t) = s.PCD(t) - (cte)(s.AGE)$$

where $s.AGE$ is the age of the meristem since it was created.

Supplementary figures

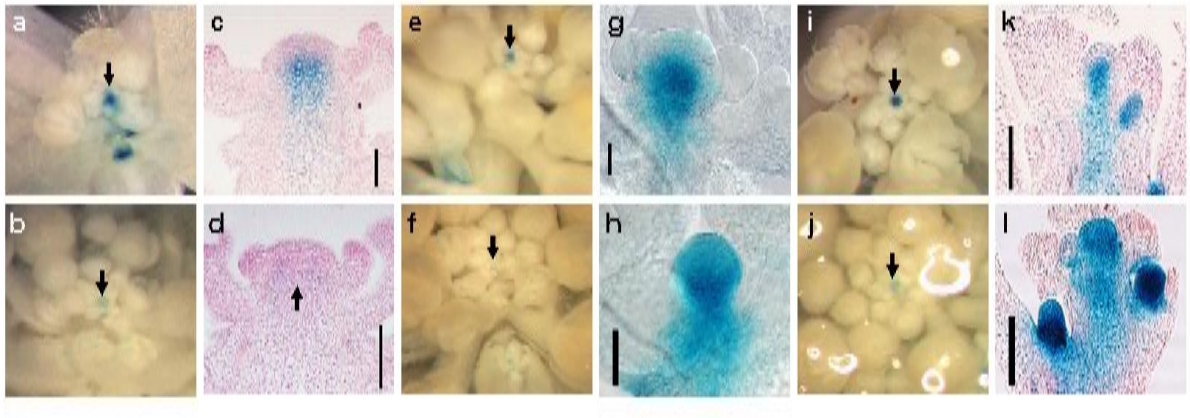


Figure S1. The photoperiod flowering pathway upregulates TFL1p:GUS activity in the shoot apex. (a-b), TFL1p:GUS activity in representative shoot apices grown under long-day (a) or short-day (b) photoperiod. **(c-d)**, longitudinal sections through the apices shown on the left. **(e-f)**, TFL1p:GUS activity in representative WT (e) and *co-3* (f) shoot apices at a similar developmental stage. **(g-h)**, TFL1p:GUS activity in cleared WT (g) and *35Sp:CO* (h) shoot apices. **(i-j)**, TFL1p:GUS activity in representative WT (i) and *ft-3* (j) shoot apices at a similar developmental stage. **(k-l)**, TFL1p:GUS activity in longitudinal sections through representative WT (k) and *35Sp:FT* (l) shoot apices at a similar developmental stage. Arrows mark the SAM region. Scale bars in (c), (d), (k) and (l), 100 µm. Scale bars in (g) and (h), 40 µm.

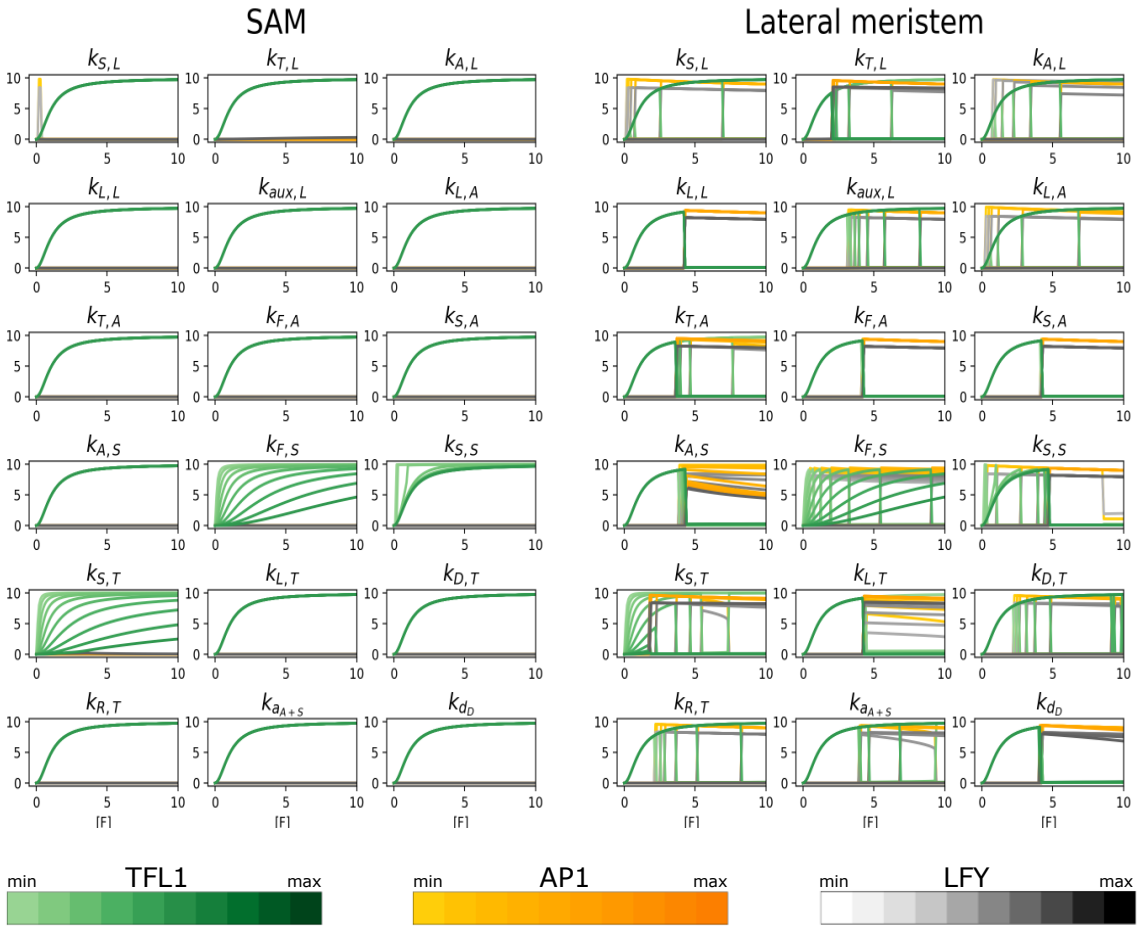


Figure S2: Robustness of the SALT model

Steady state of the SALT GRN (y axis) at different F values (x axis) using 10 different values of the parameter indicated above each graph (see Modeling Methods section), while the other parameters remain fixed. The first three columns of graphs show the analysis in SAM conditions (i.e., without auxin), and the next three columns show the analysis performed in lateral meristem conditions (i.e., with auxin). The value of the parameters changes in regular intervals from 0.001 (light green, yellow and grey colored curves) to double its optimized value (dark green, orange and dark grey colored curves). Green, yellow and red curves correspond to TFL1, AP1, and LFY value, respectively. The same qualitative behavior is observed independently of changes in the value of the parameters.

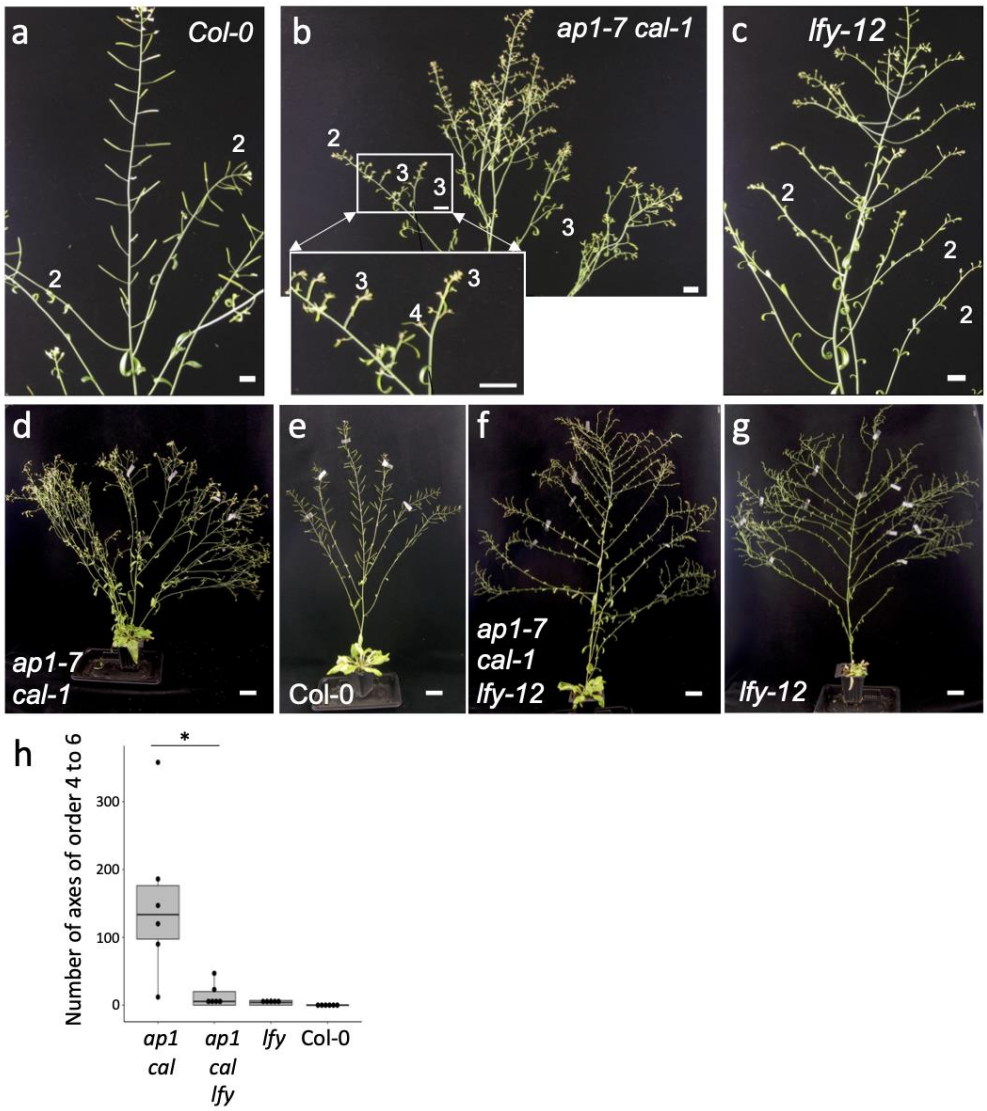
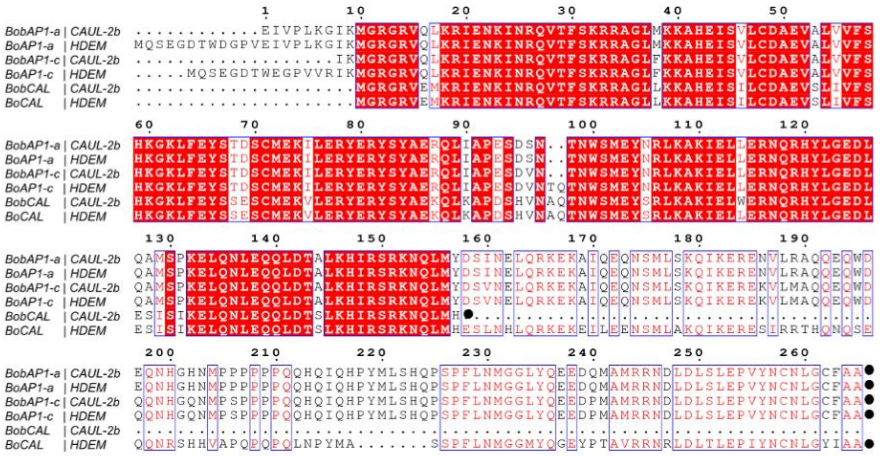


Figure S3: Genetic control of lateral axes development: Lateral axes of higher order develop in *ap1 cal* , in a *LFY* dependent manner, compared to wild-type or *lfy* plants.

The main inflorescence stem is order 1, a lateral axis growing on order n stem has order n+1. Numbers on the pictures below indicate order of the closest inflorescence.

(a-c) Inflorescences of six-week-old plants grown in long days. Whereas in these growth conditions, inflorescences of orders >2 are rarely observed in Col-0 (a) or *lfy* mutant (c), several of them are visible in *ap1 cal* mutant (b). Scale bar = 1 cm. (d-f) The highly branched structure of *ap1 cal (d)* compared to wild type Col-0 (e) is lost in *ap1 cal lfy* triple mutant (f), that have an architecture very close to the *lfy* single mutant (g). Plants were grown for 2 months in long-day conditions. Scale bar = 5 cm. Scoring axis of order ≥4 per plant (h) shows that the *ap1 cal lfy* plants behaves differently from *ap1 cal* plants (Wilcoxon test, * p-value 0.0099)

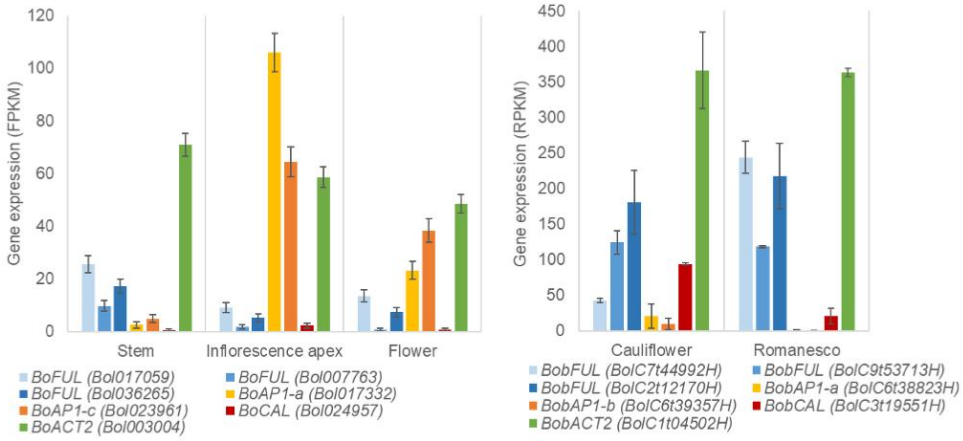
a


BobAP1-a | CAUL-2b
BoAP1-a | HDEM
BobAP1-c | CAUL-2b
BoAP1-c | HDEM
BobCAL | CAUL-2b
BoCAL | HDEM

b

OBS-Caul-1
OBS-Caul-2
OBS-Caul-3
OBS-Caul-4
OBS-Caul-5
OBS-Caul-6
CAUL-1
CAUL-2a
CAUL-2b
ROM-1
ROM-2a
ROM-2b
BobAP1-a (BolC6t38823H) 0 0 0 0 0 0 11 10 41 0 1 1
BobAP1-b (BolC2t08653H) 0 0 0 0 0 0 0 0 0 0 0 0
BobAP1-c (BolC6t39357H) 0 0 0 0 0 0 6 5 19 0 0 0
BobCAL (BolC3t19551H) 2 16 3 29 83 13 95 92 96 9 30 25
BobFUL-a (BolC2t12170H) 43 42 40 75 154 123 227 178 137 267 209 176
BobFUL-b (BolC7t44992H) 84 64 57 85 86 93 45 44 40 220 247 264
BobFUL-c (BolC9t53713H) 104 107 69 118 137 110 135 133 105 117 120 118
BobACT2 (BolC1t04502H) 516 400 386 382 567 771 419 311 368 362 358 369
Gene expression (RPKM)
700
600
500
400
300
200
100
0

c


Gene expression (FPKM)
Stem
Inflorescence apex
Flower
BoFUL (Bol017059)
BoFUL (Bol036265)
BoAP1-c (Bol023961)
BoACT2 (Bol003004)
BoFUL (Bol007763)
BoAP1-a (Bol017332)
BoCAL (Bol024957)
Gene expression (RPKM)
Cauliflower
Romanesco
BobFUL (BolC7t44992H)
BobFUL (BolC2t12170H)
BobAP1-b (BolC6t39357H)
BobACT2 (BolC1t04502H)
BobFUL (BolC9t53713H)
BobAP1-a (BolC6t38823H)
BobCAL (BolC3t19551H)

d

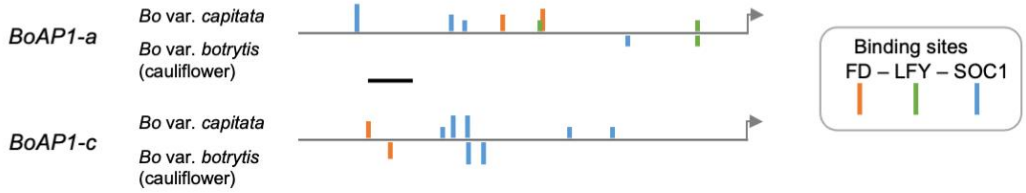

BoAP1-a
Bo var. capitata
Bo var. botrytis
(cauliflower)
BoAP1-c
Bo var. capitata
Bo var. botrytis
(cauliflower)
Binding sites
FD – LFY – SOC1

Figure S4: Sequence and expression of *BoAP1* and *BoCAL* genes.

(a) Sequence alignment of the BoAP1-a, BoAP1-c and BoCAL proteins from *Brassica oleracea (Bo)* and *Brassica oleracea* var. *botrytis (Bob)*. Sequences of the proteins were translated from the *Brassica oleracea* reference genome (*63*) or from the gene sequence assembled from reads obtained from RNA-seq of cauliflower curds (sample CAUL-2b). The mutation identified in BoCAL protein is the same as previously published (*7*)(b). Expression levels of *BobAP1*, *BobCAL* and their close homologs *FRUITFUL* (*BobFUL)* based on RNA-seq. For each gene, the normalized number of reads (RPKM) is shown as a value and color coded according to the color scale. OBS are cauliflowers harvested in the field of the Brittany Seed Observatory, CAUL and ROM samples are issued from two mature curds bought on the market (2a and 2b are different samples from curd 2). *BobAP1* genes expression is undetectable in growing cauliflowers (OBS samples) and weak in mature cauliflowers as compared to *BoFUL* genes. (c) Expression levels of genes *BoAP1*, *BoCAL* and *BoFUL* gene in *Brassica oleracea* tissues (stem, inflorescence and flowers, left panel) and the curds of the *Botrytis* varieties (cauliflower and Romanesco). Error bars represent the standard deviation between replicates. This comparison shows that *BoAP1-a* and *BoAP1-c* have a higher expression level than *BoFUL* in inflorescence and flower tissue whereas their expression is much lower than Bo*FUL's* in curds (right panel). *BobACT2* is an *ACTIN2* gene used as a reference. (d) Analysis of *BoAP1-a/c* and *BobAP1-a/c* promoter sequences using models (position weight matrices) detecting binding sites for 3 known *AP1* regulators from Arabidopsis (SOC1, FD and LFY)(*66*, *67*). Sites are denoted by bars above the line for cabbage and below for cauliflowers and with height dependent on the score obtained with the binding models. Scale bar is 100 bp.

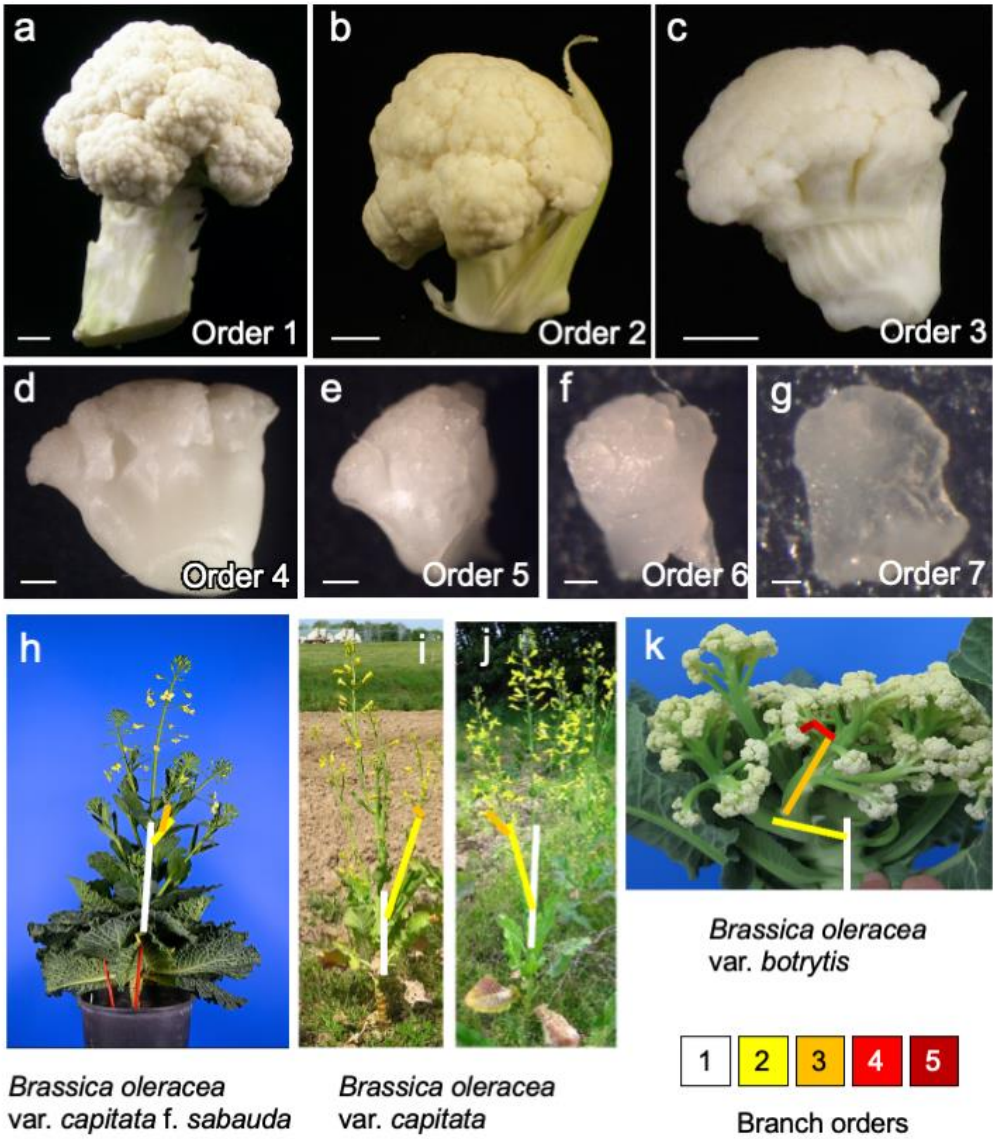


Figure S5: **Cauliflowers grow axes of higher order than cabbages**

Dissected cauliflower reveals axes of successive orders from order 1 to 7 (b-g). Growing cabbages (h-j) and an elongating cauliflower curd (k). Cabbages (h-j) show 3 orders, rarely 4, whereas cauliflowers show up to 7. Scale bars: 2 cm (a), 0.5 cm (b, c), 1 mm (d), 500 µm (e), 200 µm (f), 50 µm (g).

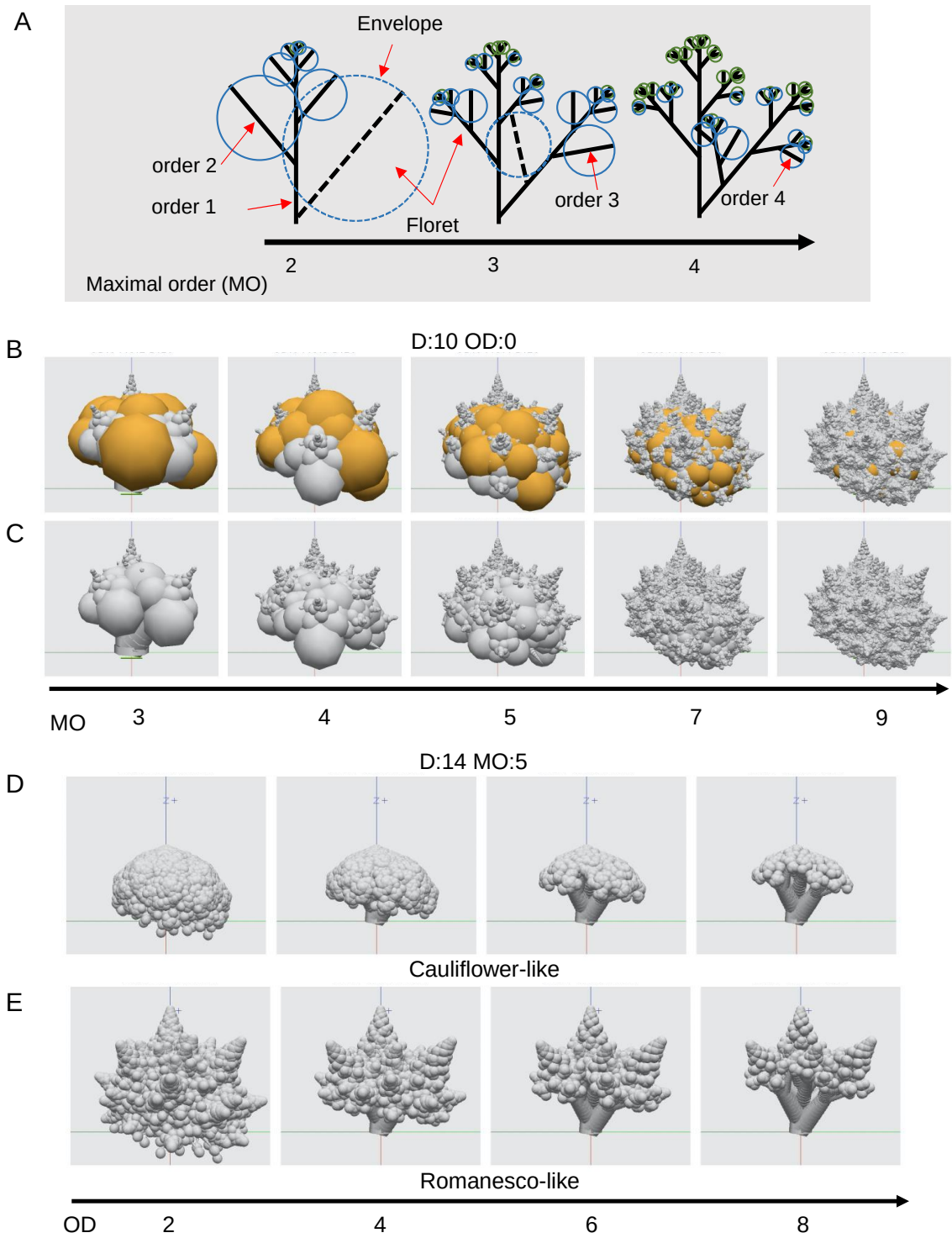


Figure S6: Overview of the geometric model characteristics and output

Labels: D = number of simulation Days, MO = Maximum Order, OD = Onset Delay.

A: Strategy to display envelopes at different maximum orders. Green circles = meristems, blue circles = floret envelopes. For aging structures, a heuristic rule was designed to remove poor envelope approximations (dashed elements). Only structures at MO are removed, meaning that structures removed at MO=k do appear at MO=k+1, thus making a sequence of increasingly precise approximations of the curd structure, and where the volume of removed elements vanishes with increasing maximum orders.

B-C: Effect of changing MO from 3-4-5-7-9 in romanesco-like curds. B. All envelopes are kept. Orange envelopes are older florets that poorly approximate the structure of the floret. C. Envelopes of old structures (in orange in B) are removed. Note that the sequence of shapes converges when MO increases, similarly to pre-fractal sequences that converge toward limit fractal structures and that at each MO, the structure with removed envelopes (C) is a better approximation of the most detailed simulation (MO=9) than when the envelopes are kept (B).

D-E: Curd simulations using meristems of fixed size and no elements removed at different OD (D: cauliflower-like, E: romanesco-like). Note that increasing OD (2-4-6-8) increasingly reduces the total number of simulated meristems. Simulations run over 14 days, MO=5.

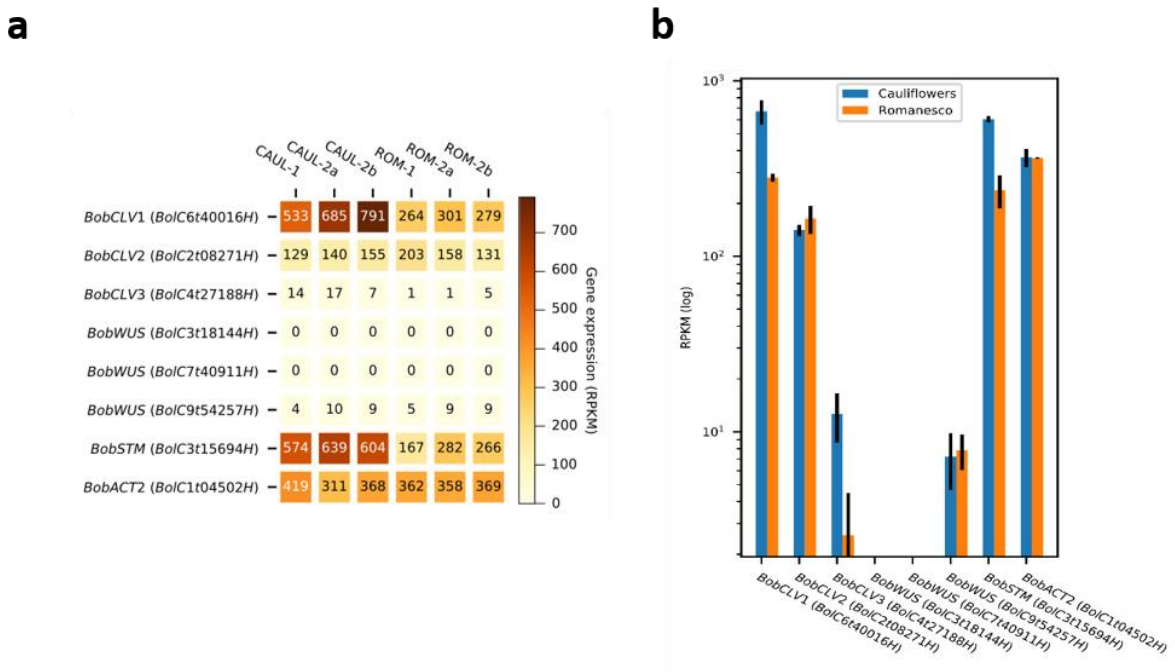


Figure S7: **Differential expression of selected meristem genes between Cauliflower and Romanesco curds.** Heatmap (a) and quantification (b) showing the expression of genes involved in meristem homeostasis in cauliflower and Romanesco curds. *CLV3* is expressed at lower level in Romanesco as compared to cauliflower (p = 0.0765; Mann & Whitney test). *CLAVATA1* and *SHOOTMERISTEMLESS* (*STM)* have lower read numbers in Romanesco but the difference is barely significant (p = 0.1; Mann & Whitney test). Error bars represent the standard deviation between replicates.

Movie Captions

Simulations of the realistic model of *Arabidopsis thaliana* coupling architecture development with the GRN model (arabidopsis-grn.lpy file)

- **Movie 1:** growth of a wild-type plant : no gene mutated
- **Movie 2:** growth of *lfy* mutant: *LFY* gene off
- **Movie 3:** growth of *ap1/cal* mutant: *AP1/CAL* gene off

Simulations of the simple geometric model (geometric-model-erratum.lpy file)

- **Movie 4:** growth of a cauliflower-like structure using the geometric model (geometric-model-erratum.lpy), where all apices are represented by identical small spheres and no structure is removed (Flag CONST_RADIUS on). The simulation runs over 16 days with MO=4 and OD=6.

- **Movie 5:** growth of a romanesco-like structure using our geometric model (geometric-model-erratum.lpy), where all apices are represented by identical small spheres and no structure is removed (Flag CONST_RADIUS on). This simulation is obtained by simply replacing the constant plastochron law used in movie 4 by a decreasing plastochron. The simulation runs over 16 days with MO=4 and OD=6.

Table S1. List of interactions used in the GRN.

Regulator	Type of regulation	References
AP1/CAL regulations		
LFY	**Positive direct**	(*15*, *41*, *67*, *76–80*)
TFL1	**Negative direct (in complex with FD)**	(*26*, *29*, *81–83*)
SAX	**Positive direct**	(*21*, *41*)
F	**Positive (direct in complex with FD)**	(*84–89*)
LFY regulations		
AP1/CAL	**Positive direct**	(*6*, *82*, *90*, *91*)
TFL1	**Negative direct**	(*17*, *26*, *29*, *83*)
SAX	**Positive direct**	(*21*, *41*, *92–94*)
Auxin	**Positive via MP/ARF5**	(*95*, *96*)
LFY	**Negative direct**	(*15*, *67*, *76*, *96*, *97*)
TFL1 regulations		
AP1/CAL	**Negative direct**	(*6*, *25*, *82*, *90*, *91*)
SAX	**Positive direct**	This study, (*21*)
SAX	**Negative direct**	(*21*, *25*)
LFY	**Positive direct**	(*15*, *67*, *76*, *98*)
SAX regulations		
AP1	**Negative direct**	(*9*, *21*, *91*, *99*, *100*)
SAX	**Positive direct**	(*21*, *28*, *42*, *92*, *94*, *101*, *102*)
F	**Positive direct**	(*18*, *19*, *21*, *44*, *89*, *93*, *103–108*)

Table S2. Parameter values used in the SALT GRN.
All $k_{X,Y}$ are in nM t^{-1} units. Time (t) is in arbitrary units.

$k_{L,A}$	1
$k_{T,A}$	1
$k_{F,A}$	50
$k_{S,A}$	15
$k_{A,S}$	4
$k_{F,S}$	10
$k_{S,S}$	20
$k_{S,T}$	1
$k_{L,T}$	100
$k_{D,T}$	1
$k_{R,T}$	0.1
$k_{S,L}$	6
$k_{A,L}$	4
$k_{L,L}$	100
$k_{T,L}$	5
$k_{aux,L}$	1
$k_{a_{S+A}}$	0.1
k_{d_D}	0.01
δ	0.1
n	2
L_{th}	1

Table S3. Feedback and feedforward loops in the SALT GRN.

Feedback loops and their sign	Coherent and Incoherent FFLs (ignoring SAX)
▪ Mutual activation AP1←→LFY (>0) ▪ Mutual repression AP1⊢⊣TFL1 (>0) ▪ Positive 3-loop LFY→AP1⊣TFL1⊣LFY (>0) ▪ Self repression LFY ⊢⊣LFY (<0) ▪ Negative 2-loop LFY⊢→TFL1 (<0) ▪ Negative 3-loop LFY→TFL1⊣AP1→LFY (<0)	▪ CFFL, type 2: TFL1⊣AP1→LFY / TFL1⊣LFY ▪ CFFL, type 2: TFL1⊣LFY→AP1/ TFL1⊣AP1 ▪ CFFL, type 4: AP1⊣TFL1⊣LFY/ AP1→LFY ▪ IFFL, type 3: AP1→LFY→TFL1/ AP1⊣TFL1 ▪ IFFL, type 1: LFY→TFL1⊣AP1/ LFY→AP1 ▪ IFFL, type 1: LFY→AP1⊣TFL1/ LFY→TFL1 ▪ SAX->LFY->AP1 / SAX->AP1; CFFL type 1 ▪ SAX->LFY->TLF1 / SAX->TFL1; CFFL type 1 ▪ SAX->AP1->LFY/ SAX->LFY; CFFL type 1 ▪ SAX->AP1-|TLF1/ SAX->TFL1; IFFL type 1 ▪ SAX->TFL1-|LFY / SAX->LFY; IFFL type 1 ▪ SAX->TFL1-|AP1 / SAX->AP1; IFFL type 1

References and Notes

1. C. Godin, C. Golé, S. Douady, Phyllotaxis as geometric canalization during plant development. *Development* **147**, dev165878 (2020). doi:10.1242/dev.165878 Medline

2. C. F. Quiros, M. W. Farnham, "The genetics of *Brassica oleracea*," in *Genetics and Genomics of the Brassicaceae*, R. Schmidt, I. Bancroft, Eds. (Springer, 2011), pp. 261–289.

3. D. V. Duclos, T. Björkman, Meristem identity gene expression during curd proliferation and flower initiation in *Brassica oleracea*. *J. Exp. Bot.* **59**, 421–433 (2008). doi:10.1093/jxb/erm327 Medline

4. L. B. Smith, G. J. King, The distribution of BoCAL-a alleles in *Brassica oleracea* is consistent with a genetic model for curd development and domestication of the cauliflower. *Mol. Breed.* **6**, 603–613 (2000). doi:10.1023/A:1011370525688

5. N. Guo, S. Wang, L. Gao, Y. Liu, X. Wang, E. Lai, M. Duan, G. Wang, J. Li, M. Yang, M. Zong, S. Han, Y. Pei, T. Borm, H. Sun, L. Miao, D. Liu, F. Yu, W. Zhang, H. Ji, C. Zhu, Y. Xu, G. Bonnema, J. Li, Z. Fei, F. Liu, Genome sequencing sheds light on the contribution of structural variants to *Brassica oleracea* diversification. *BMC Biol.* **19**, 93 (2021). doi:10.1186/s12915-021-01031-2 Medline

6. J. L. Bowman, J. Alvarez, D. Weigel, E. M. Meyerowitz, D. R. Smyth, Control of flower development in *Arabidopsis thaliana* by APETALA1 and interacting genes. *Development* **119**, 721–743 (1993). doi:10.1242/dev.119.3.721

7. S. A. Kempin, B. Savidge, M. F. Yanofsky, Molecular basis of the cauliflower phenotype in *Arabidopsis*. *Science* **267**, 522–525 (1995). doi:10.1126/science.7824951 Medline

8. G. Denay, H. Chahtane, G. Tichtinsky, F. Parcy, A flower is born: An update on *Arabidopsis* floral meristem formation. *Curr. Opin. Plant Biol.* **35**, 15–22 (2017). doi:10.1016/j.pbi.2016.09.003 Medline

9. A. Pajoro, S. Biewers, E. Dougali, F. Leal Valentim, M. A. Mendes, A. Porri, G. Coupland, Y. Van de Peer, A. D. van Dijk, L. Colombo, B. Davies, G. C. Angenent, The (r)evolution of gene regulatory networks controlling *Arabidopsis* plant reproduction: A two-decade history. *J. Exp. Bot.* **65**, 4731–4745 (2014). doi:10.1093/jxb/eru233 Medline

10. B. Thomson, F. Wellmer, Molecular regulation of flower development. *Curr. Top. Dev. Biol.* **131**, 185–210 (2019). doi:10.1016/bs.ctdb.2018.11.007 Medline

11. K. E. Jaeger, N. Pullen, S. Lamzin, R. J. Morris, P. A. Wigge, Interlocking feedback loops govern the dynamic behavior of the floral transition in *Arabidopsis*. *Plant Cell* **25**, 820–833 (2013). doi:10.1105/tpc.113.109355 Medline

12. C. Espinosa-Soto, P. Padilla-Longoria, E. R. Alvarez-Buylla, A gene regulatory network model for cell-fate determination during *Arabidopsis thaliana* flower development that is robust and recovers experimental gene expression profiles. *Plant Cell* **16**, 2923–2939 (2004). doi:10.1105/tpc.104.021725 Medline

13. F. Leal Valentim, Sv. Mourik, D. Posé, M. C. Kim, M. Schmid, R. C. van Ham, M. Busscher, G. F. Sanchez-Perez, J. Molenaar, G. C. Angenent, R. G. Immink, A. D. van Dijk, A quantitative and dynamic model of the *Arabidopsis* flowering time gene

regulatory network. *PLOS ONE* **10**, e0116973 (2015). doi:10.1371/journal.pone.0116973 Medline

14. P. Prusinkiewicz, Y. Erasmus, B. Lane, L. D. Harder, E. Coen, Evolution and development of inflorescence architectures. *Science* **316**, 1452–1456 (2007). doi:10.1126/science.1140429 Medline

15. K. Goslin, B. Zheng, A. Serrano-Mislata, L. Rae, P. T. Ryan, K. Kwaśniewska, B. Thomson, D. S. Ó'Maoiléidigh, F. Madueño, F. Wellmer, E. Graciet, Transcription Factor Interplay between LEAFY and APETALA1/CAULIFLOWER during Floral Initiation. *Plant Physiol.* **174**, 1097–1109 (2017). doi:10.1104/pp.17.00098 Medline

16. C. Ferrándiz, Q. Gu, R. Martienssen, M. F. Yanofsky, Redundant regulation of meristem identity and plant architecture by FRUITFULL, APETALA1 and CAULIFLOWER. *Development* **127**, 725–734 (2000). doi:10.1242/dev.127.4.725 Medline

17. D. Bradley, O. Ratcliffe, C. Vincent, R. Carpenter, E. Coen, Inflorescence commitment and architecture in *Arabidopsis*. *Science* **275**, 80–83 (1997). doi:10.1126/science.275.5296.80 Medline

18. X. Hou, J. Zhou, C. Liu, L. Liu, L. Shen, H. Yu, Nuclear factor Y-mediated H3K27me3 demethylation of the SOC1 locus orchestrates flowering responses of *Arabidopsis*. *Nat. Commun.* **5**, 4601 (2014). doi:10.1038/ncomms5601 Medline

19. S. K. Yoo, K. S. Chung, J. Kim, J. H. Lee, S. M. Hong, S. J. Yoo, S. Y. Yoo, J. S. Lee, J. H. Ahn, *CONSTANS* activates *SUPPRESSOR OF OVEREXPRESSION OF CONSTANS 1* through *FLOWERING LOCUS T* to promote flowering in *Arabidopsis*. *Plant Physiol.* **139**, 770–778 (2005). doi:10.1104/pp.105.066928 Medline

20. A. Serrano-Mislata, P. Fernández-Nohales, M. J. Doménech, Y. Hanzawa, D. Bradley, F. Madueño, Separate elements of the TERMINAL FLOWER 1 cis-regulatory region integrate pathways to control flowering time and shoot meristem identity. *Development* **143**, 3315–3327 (2016). Medline

21. R. V. Pérez-Ruiz, B. García-Ponce, N. Marsch-Martínez, Y. Ugartechea-Chirino, M. Villajuana-Bonequi, S. de Folter, E. Azpeitia, J. Dávila-Velderrain, D. Cruz-Sánchez, A. Garay-Arroyo, Mde. L. Sánchez, J. M. Estévez-Palmas, E. R. Álvarez-Buylla, XAANTAL2 (AGL14) is an important component of the complex gene regulatory network that underlies *Arabidopsis* shoot apical meristem transitions. *Mol. Plant* **8**, 796–813 (2015). doi:10.1016/j.molp.2015.01.017 Medline

22. D. Reinhardt, E.-R. Pesce, P. Stieger, T. Mandel, K. Baltensperger, M. Bennett, J. Traas, J. Friml, C. Kuhlemeier, Regulation of phyllotaxis by polar auxin transport. *Nature* **426**, 255–260 (2003). doi:10.1038/nature02081 Medline

23. P. A. Wigge, FT, a mobile developmental signal in plants. *Curr. Biol.* **21**, R374–R378 (2011). doi:10.1016/j.cub.2011.03.038 Medline

24. J. Putterill, E. Varkonyi-Gasic, FT and florigen long-distance flowering control in plants. *Curr. Opin. Plant Biol.* **33**, 77–82 (2016). doi:10.1016/j.pbi.2016.06.008 Medline

25. C. Liu, Z. W. N. Teo, Y. Bi, S. Song, W. Xi, X. Yang, Z. Yin, H. Yu, A conserved genetic pathway determines inflorescence architecture in *Arabidopsis* and rice. *Dev. Cell* **24**, 612–622 (2013). doi:10.1016/j.devcel.2013.02.013 Medline

26. O. J. Ratcliffe, I. Amaya, C. A. Vincent, S. Rothstein, R. Carpenter, E. S. Coen, D. J. Bradley, A common mechanism controls the life cycle and architecture of plants. *Development* **125**, 1609–1615 (1998). doi:10.1242/dev.125.9.1609 Medline

27. D. Weigel, J. Alvarez, D. R. Smyth, M. F. Yanofsky, E. M. Meyerowitz, LEAFY controls floral meristem identity in *Arabidopsis*. *Cell* **69**, 843–859 (1992). doi:10.1016/0092-8674(92)90295-N Medline

28. S. D. Michaels, G. Ditta, C. Gustafson-Brown, S. Pelaz, M. Yanofsky, R. M. Amasino, *AGL24* acts as a promoter of flowering in *Arabidopsis* and is positively regulated by vernalization. *Plant J.* **33**, 867–874 (2003). doi:10.1046/j.1365-313X.2003.01671.x Medline

29. Y. Zhu, S. Klasfeld, C. W. Jeong, R. Jin, K. Goto, N. Yamaguchi, D. Wagner, TERMINAL FLOWER 1-FD complex target genes and competition with FLOWERING LOCUS T. *Nat. Commun.* **11**, 5118 (2020). doi:10.1038/s41467-020-18782-1 Medline

30. L. Conti, D. Bradley, TERMINAL FLOWER1 is a mobile signal controlling *Arabidopsis* architecture. *Plant Cell* **19**, 767–778 (2007). doi:10.1105/tpc.106.049767 Medline

31. H. Chahtane, G. Vachon, M. Le Masson, E. Thévenon, S. Périgon, N. Mihajlovic, A. Kalinina, R. Michard, E. Moyroud, M. Monniaux, C. Sayou, V. Grbic, F. Parcy, G. Tichtinsky, A variant of LEAFY reveals its capacity to stimulate meristem development by inducing *RAX1*. *Plant J.* **74**, 678–689 (2013). doi:10.1111/tpj.12156 Medline

32. D. Weigel, O. Nilsson, A developmental switch sufficient for flower initiation in diverse plants. *Nature* **377**, 495–500 (1995). doi:10.1038/377495a0 Medline

33. X. Sun, J. Bucher, Y. Ji, A. D. J. van Dijk, R. G. H. Immink, G. Bonnema, Effect of ambient temperature fluctuation on the timing of the transition to the generative stage in cauliflower. *Environ. Exp. Bot.* **155**, 742–750 (2018). doi:10.1016/j.envexpbot.2018.06.013

34. M. Kieffer, M. P. Fuller, A. J. Jellings, Explaining curd and spear geometry in broccoli, cauliflower and 'romanesco': Quantitative variation in activity of primary meristems. *Planta* **206**, 34–43 (1998). doi:10.1007/s004250050371

35. S. Douady, Y. Couder, Phyllotaxis as a dynamical self organizing process part I: The spiral modes resulting from time-periodic iterations. *J. Theor. Biol.* **178**, 255–273 (1996). doi:10.1006/jtbi.1996.0024

36. Y. Refahi, G. Brunoud, E. Farcot, A. Jean-Marie, M. Pulkkinen, T. Vernoux, C. Godin, A stochastic multicellular model identifies biological watermarks from disorders in self-organized patterns of phyllotaxis. *eLife* **5**, e14093 (2016). doi:10.7554/eLife.14093 Medline

37. J. C. Fletcher, U. Brand, M. P. Running, R. Simon, E. M. Meyerowitz, Signaling of cell fate decisions by CLAVATA3 in *Arabidopsis* shoot meristems. *Science* **283**, 1911–1914 (1999). doi:10.1126/science.283.5409.1911 Medline

38. M. Kitagawa, D. Jackson, Control of meristem size. *Annu. Rev. Plant Biol.* **70**, 269–291 (2019). doi:10.1146/annurev-arplant-042817-040549 Medline

39. L. E. Watts, Selection for quality in marrow-stem kale. *Euphytica* **15**, 224–228 (1966). doi:10.1007/BF00022327

40. A. Maizel, D. Weigel, Temporally and spatially controlled induction of gene expression in *Arabidopsis thaliana*. *Plant J.* **38**, 164–171 (2004). doi:10.1111/j.1365-313X.2004.02027.x Medline

41. V. Grandi, V. Gregis, M. M. Kater, Uncovering genetic and molecular interactions among floral meristem identity genes in *Arabidopsis thaliana*. *Plant J.* **69**, 881–893 (2012). doi:10.1111/j.1365-313X.2011.04840.x Medline

42. R. G. H. Immink, D. Posé, S. Ferrario, F. Ott, K. Kaufmann, F. L. Valentim, S. de Folter, F. van der Wal, A. D. J. van Dijk, M. Schmid, G. C. Angenent, Characterization of SOC1's central role in flowering by the identification of its upstream and downstream regulators. *Plant Physiol.* **160**, 433–449 (2012). doi:10.1104/pp.112.202614 Medline

43. I. Kardailsky, V. K. Shukla, J. H. Ahn, N. Dagenais, S. K. Christensen, J. T. Nguyen, J. Chory, M. J. Harrison, D. Weigel, Activation tagging of the floral inducer FT. *Science* **286**, 1962–1965 (1999). doi:10.1126/science.286.5446.1962 Medline

44. H. Lee, S. S. Suh, E. Park, E. Cho, J. H. Ahn, S. G. Kim, J. S. Lee, Y. M. Kwon, I. Lee, The AGAMOUS-LIKE 20 MADS domain protein integrates floral inductive pathways in *Arabidopsis*. *Genes Dev.* **14**, 2366–2376 (2000). doi:10.1101/gad.813600 Medline

45. V. Gregis, A. Sessa, C. Dorca-Fornell, M. M. Kater, The *Arabidopsis* floral meristem identity genes AP1, AGL24 and SVP directly repress class B and C floral homeotic genes. *Plant J.* **60**, 626–637 (2009). doi:10.1111/j.1365-313X.2009.03985.x Medline

46. H. Onouchi, M. I. Igeño, C. Périlleux, K. Graves, G. Coupland, Mutagenesis of plants overexpressing *CONSTANS* demonstrates novel interactions among *Arabidopsis* flowering-time genes. *Plant Cell* **12**, 885–900 (2000). doi:10.1105/tpc.12.6.885 Medline

47. M. Koornneef, C. J. Hanhart, J. H. van der Veen, A genetic and physiological analysis of late flowering mutants in *Arabidopsis thaliana*. *Mol. Gen. Genet.* **229**, 57–66 (1991). doi:10.1007/BF00264213 Medline

48. F. Wellmer, M. Alves-Ferreira, A. Dubois, J. L. Riechmann, E. M. Meyerowitz, Genome-wide analysis of gene expression during early *Arabidopsis* flower development. *PLOS Genet.* **2**, e117 (2006). doi:10.1371/journal.pgen.0020117 Medline

49. R. A. Jefferson, T. A. Kavanagh, M. W. Bevan, GUS fusions: Beta-glucuronidase as a sensitive and versatile gene fusion marker in higher plants. *EMBO J.* **6**, 3901–3907 (1987). doi:10.1002/j.1460-2075.1987.tb02730.x Medline

50. N. Bechtold, J. Ellis, G. Pelletier, C. R. Acad. Sci. Paris. *Life Sci.* **316**, 1194–1199 (1993).

51. N. Bechtold, D. Bouchez, "In planta *Agrobacterium*-mediated transformation of adult *Arabidopsis thaliana* plants by vacuum infiltration," in *Gene Transfer to Plants*, I. Potrykus, G. Spangenberg, Eds. (Springer, 1995) pp. 19–23.

52. M. Trigueros, M. Navarrete-Gómez, S. Sato, S. K. Christensen, S. Pelaz, D. Weigel, M. F. Yanofsky, C. Ferrándiz, The *NGATHA* genes direct style development in the *Arabidopsis* gynoecium. *Plant Cell* **21**, 1394–1409 (2009). doi:10.1105/tpc.109.065508 Medline

53. I. Mitsuhara, M. Ugaki, H. Hirochika, M. Ohshima, T. Murakami, Y. Gotoh, Y. Katayose, S. Nakamura, R. Honkura, S. Nishimiya, K. Ueno, A. Mochizuki, H. Tanimoto, H. Tsugawa, Y. Otsuki, Y. Ohashi, Efficient promoter cassettes for enhanced expression of foreign genes in dicotyledonous and monocotyledonous plants. *Plant Cell Physiol.* **37**, 49–59 (1996). doi:10.1093/oxfordjournals.pcp.a028913 Medline

54. R. P. Hellens, A. C. Allan, E. N. Friel, K. Bolitho, K. Grafton, M. D. Templeton, S. Karunairetnam, A. P. Gleave, W. A. Laing, Transient expression vectors for functional genomics, quantification of promoter activity and RNA silencing in plants. *Plant Methods* **1**, 13 (2005). doi:10.1186/1746-4811-1-13 Medline

55. Z. Feng, B. Zhang, W. Ding, X. Liu, D.-L. Yang, P. Wei, F. Cao, S. Zhu, F. Zhang, Y. Mao, J.-K. Zhu, Efficient genome editing in plants using a CRISPR/Cas system. *Cell Res.* **23**, 1229–1232 (2013). doi:10.1038/cr.2013.114 Medline

56. W. Yan, D. Chen, K. Kaufmann, Efficient multiplex mutagenesis by RNA-guided Cas9 and its use in the characterization of regulatory elements in the AGAMOUS gene. *Plant Methods* **12**, 23 (2016). doi:10.1186/s13007-016-0125-7 Medline

57. S. Bensmihen, A. To, G. Lambert, T. Kroj, J. Giraudat, F. Parcy, Analysis of an activated *ABI5* allele using a new selection method for transgenic *Arabidopsis* seeds. *FEBS Lett.* **561**, 127–131 (2004). doi:10.1016/S0014-5793(04)00148-6 Medline

58. S. J. Clough, A. F. Bent, Floral dip: A simplified method for *Agrobacterium*-mediated transformation of *Arabidopsis thaliana*. *Plant J.* **16**, 735–743 (1998). doi:10.1046/j.1365-313x.1998.00343.x Medline

59. N. Prunet, K. Duncan, Imaging flowers: A guide to current microscopy and tomography techniques to study flower development. *J. Exp. Bot.* **71**, 2898–2909 (2020). doi:10.1093/jxb/eraa094 Medline

60. A. Sessions, D. Weigel, M. F. Yanofsky, The *Arabidopsis thaliana* MERISTEM LAYER 1 promoter specifies epidermal expression in meristems and young primordia. *Plant J.* **20**, 259–263 (1999). doi:10.1046/j.1365-313x.1999.00594.x Medline

61. D. Weigel, J. Glazebrook, *Arabidopsis: A Laboratory Manual* (Cold Spring Harbor Laboratory Press, 2002).

62. C. Dorca-Fornell, V. Gregis, V. Grandi, G. Coupland, L. Colombo, M. M. Kater, The *Arabidopsis* SOC1-like genes AGL42, AGL71 and AGL72 promote flowering in the shoot apical and axillary meristems. *Plant J.* **67**, 1006–1017 (2011). doi:10.1111/j.1365-313X.2011.04653.x Medline

63. C. Belser, B. Istace, E. Denis, M. Dubarry, F.-C. Baurens, C. Falentin, M. Genete, W. Berrabah, A.-M. Chèvre, R. Delourme, G. Deniot, F. Denoeud, P. Duffé, S. Engelen, A. Lemainque, M. Manzanares-Dauleux, G. Martin, J. Morice, B. Noel, X. Vekemans, A. D'Hont, M. Rousseau-Gueutin, V. Barbe, C. Cruaud, P. Wincker, J.-M. Aury, Chromosome-scale assemblies of plant genomes using nanopore long reads and optical maps. *Nat. Plants* **4**, 879–887 (2018). doi:10.1038/s41477-018-0289-4 Medline

64. J. Yu, S. Tehrim, F. Zhang, C. Tong, J. Huang, X. Cheng, C. Dong, Y. Zhou, R. Qin, W. Hua, S. Liu, Genome-wide comparative analysis of NBS-encoding genes between Brassica species and *Arabidopsis thaliana*. *BMC Genomics* **15**, 3 (2014). doi:10.1186/1471-2164-15-3 Medline

65. H. Lv, Y. Wang, F. Han, J. Ji, Z. Fang, M. Zhuang, Z. Li, Y. Zhang, L. Yang, A high-quality reference genome for cabbage obtained with SMRT reveals novel genomic features and evolutionary characteristics. *Sci. Rep.* **10**, 12394 (2020). doi:10.1038/s41598-020-69389-x Medline

66. O. Fornes, J. A. Castro-Mondragon, A. Khan, R. van der Lee, X. Zhang, P. A. Richmond, B. P. Modi, S. Correard, M. Gheorghe, D. Baranašić, W. Santana-Garcia, G. Tan, J. Chèneby, B. Ballester, F. Parcy, A. Sandelin, B. Lenhard, W. W. Wasserman, A. Mathelier, JASPAR 2020: Update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.* **48** (D1), D87–D92 (2020). Medline

67. E. Moyroud, E. G. Minguet, F. Ott, L. Yant, D. Posé, M. Monniaux, S. Blanchet, O. Bastien, E. Thévenon, D. Weigel, M. Schmid, F. Parcy, Prediction of regulatory interactions from genome sequences using a biophysical model for the *Arabidopsis* LEAFY transcription factor. *Plant Cell* **23**, 1293–1306 (2011). doi:10.1105/tpc.111.083329 Medline

68. B. H. Toyama, M. W. Hetzer, Protein homeostasis: Live long, won't prosper. *Nat. Rev. Mol. Cell Biol.* **14**, 55–61 (2013). doi:10.1038/nrm3496 Medline

69. A. Jolma, Y. Yin, K. R. Nitta, K. Dave, A. Popov, M. Taipale, M. Enge, T. Kivioja, E. Morgunova, J. Taipale, DNA-dependent formation of transcription factor pairs alters their binding specificity. *Nature* **527**, 384–388 (2015). doi:10.1038/nature15518 Medline

70. S. Legewie, H. Herzel, H. V. Westerhoff, N. Blüthgen, Recurrent design patterns in the feedback regulation of the mammalian signalling network. *Mol. Syst. Biol.* **4**, 190 (2008). doi:10.1038/msb.2008.29 Medline

71. S. Belikov, O. G. Berg, Ö. Wrange, Quantification of transcription factor-DNA binding affinity in a living cell. *Nucleic Acids Res.* **44**, 3045–3058 (2016). doi:10.1093/nar/gkv1350 Medline

72. S. Mangan, U. Alon, Structure and function of the feed-forward loop network motif. *Proc. Natl. Acad. Sci. U.S.A.* **100**, 11980–11985 (2003). doi:10.1073/pnas.2133841100 Medline

73. P. Prusinkiewicz, A. Lindenmayer, *The Algorithmic Beauty of Plants* (Springer, 1990).

74. F. Boudon, C. Pradal, T. Cokelaer, P. Prusinkiewicz, C. Godin, L-py: An L-system simulation framework for modeling plant architecture development based on a dynamic language. *Front. Plant Sci.* **3**, 76 (2012). doi:10.3389/fpls.2012.00076 Medline

75. L. Mündermann, Y. Erasmus, B. Lane, E. Coen, P. Prusinkiewicz, Quantitative modeling of *Arabidopsis* development. *Plant Physiol.* **139**, 960–968 (2005). doi:10.1104/pp.105.060483 Medline

76. C. M. Winter, R. S. Austin, S. Blanvillain-Baufumé, M. A. Reback, M. Monniaux, M.-F. Wu, Y. Sang, A. Yamaguchi, N. Yamaguchi, J. E. Parker, F. Parcy, S. T. Jensen, H. Li, D. Wagner, LEAFY target genes reveal floral regulatory logic, cis motifs, and a link to

biotic stimulus response. *Dev. Cell* **20**, 430–443 (2011). doi:10.1016/j.devcel.2011.03.019 Medline

77. D. A. William, Y. Su, M. R. Smith, M. Lu, D. A. Baldwin, D. Wagner, Genomic identification of direct target genes of LEAFY. *Proc. Natl. Acad. Sci. U.S.A.* **101**, 1775–1780 (2004). doi:10.1073/pnas.0307842100 Medline

78. R. Benlloch, M. C. Kim, C. Sayou, E. Thévenon, F. Parcy, O. Nilsson, Integrating long-day flowering signals: A LEAFY binding site is essential for proper photoperiodic activation of APETALA1. *Plant J.* **67**, 1094–1102 (2011). doi:10.1111/j.1365-313X.2011.04660.x Medline

79. D. Wagner, R. W. M. Sablowski, E. M. Meyerowitz, Transcriptional activation of APETALA1 by LEAFY. *Science* **285**, 582–584 (1999). doi:10.1126/science.285.5427.582 Medline

80. F. Parcy, O. Nilsson, M. A. Busch, I. Lee, D. Weigel, A genetic framework for floral patterning. *Nature* **395**, 561–566 (1998). doi:10.1038/26903 Medline

81. S. Hanano, K. Goto, *Arabidopsis* TERMINAL FLOWER1 is involved in the regulation of flowering time and inflorescence development through transcriptional repression. *Plant Cell* **23**, 3172–3184 (2011). doi:10.1105/tpc.111.088641 Medline

82. O. J. Ratcliffe, D. J. Bradley, E. S. Coen, Separation of shoot and floral identity in *Arabidopsis*. *Development* **126**, 1109–1120 (1999). doi:10.1242/dev.126.6.1109 Medline

83. D. Goretti, M. Silvestre, S. Collani, T. Langenecker, C. Méndez, F. Madueño, M. Schmid, TERMINAL FLOWER1 Functions as a Mobile Transcriptional Cofactor in the Shoot Apical Meristem. *Plant Physiol.* **182**, 2081–2095 (2020). doi:10.1104/pp.19.00867 Medline

84. J.-H. Jung, H.-J. Lee, J. Y. Ryu, C.-M. Park, SPL3/4/5 integrate developmental aging and photoperiodic signals into the FT-FD module in *Arabidopsis* Flowering. *Mol. Plant* **9**, 1647–1659 (2016). doi:10.1016/j.molp.2016.10.014 Medline

85. P. A. Wigge, M. C. Kim, K. E. Jaeger, W. Busch, M. Schmid, J. U. Lohmann, D. Weigel, Integration of spatial and temporal information during floral induction in *Arabidopsis*. *Science* **309**, 1056–1059 (2005). doi:10.1126/science.1114358 Medline

86. S. Collani, M. Neumann, L. Yant, M. Schmid, FT modulates genome-wide DNA-binding of the bZIP transcription factor FD. *Plant Physiol.* **180**, 367–380 (2019). doi:10.1104/pp.18.01505 Medline

87. M. Abe, Y. Kobayashi, S. Yamamoto, Y. Daimon, A. Yamaguchi, Y. Ikeda, H. Ichinoki, M. Notaguchi, K. Goto, T. Araki, FD, a bZIP protein mediating signals from the floral pathway integrator FT at the shoot apex. *Science* **309**, 1052–1056 (2005). doi:10.1126/science.1115983 Medline

88. P. Teper-Bamnolker, A. Samach, The flowering integrator FT regulates *SEPALLATA3* and *FRUITFULL* accumulation in *Arabidopsis* leaves. *Plant Cell* **17**, 2661–2675 (2005). doi:10.1105/tpc.105.035766 Medline

89. M. Romera-Branchat, E. Severing, C. Pocard, H. Ohr, C. Vincent, G. Née, R. Martinez-Gallegos, S. Jang, F. Andrés, P. Madrigal, G. Coupland, Functional divergence of the

arabidopsis florigen-interacting bZIP transcription factors FD and FDP. *Cell Rep.* **31**, 107717 (2020). doi:10.1016/j.celrep.2020.107717 Medline

90. S. J. Liljegren, C. Gustafson-Brown, A. Pinyopich, G. S. Ditta, M. F. Yanofsky, Interactions among *APETALA1*, *LEAFY*, and *TERMINAL FLOWER1* specify meristem fate. *Plant Cell* **11**, 1007–1018 (1999). doi:10.1105/tpc.11.6.1007 Medline

91. K. Kaufmann, F. Wellmer, J. M. Muiño, T. Ferrier, S. E. Wuest, V. Kumar, A. Serrano-Mislata, F. Madueño, P. Krajewski, E. M. Meyerowitz, G. C. Angenent, J. L. Riechmann, Orchestration of floral initiation by APETALA1. *Science* **328**, 85–89 (2010). doi:10.1126/science.1185244 Medline

92. J. Lee, M. Oh, H. Park, I. Lee, SOC1 translocated to the nucleus by interaction with AGL24 directly regulates *leafy*. *Plant J.* **55**, 832–843 (2008). doi:10.1111/j.1365-313X.2008.03552.x Medline

93. J. Moon, H. Lee, M. Kim, I. Lee, Analysis of flowering pathway integrators in *Arabidopsis*. *Plant Cell Physiol.* **46**, 292–299 (2005). doi:10.1093/pcp/pci024 Medline

94. C. Liu, H. Chen, H. L. Er, H. M. Soo, P. P. Kumar, J.-H. Han, Y. C. Liou, H. Yu, Direct interaction of *AGL24* and *SOC1* integrates flowering signals in *Arabidopsis*. *Development* **135**, 1481–1491 (2008). doi:10.1242/dev.020255 Medline

95. N. Yamaguchi, M.-F. Wu, C. M. Winter, M. C. Berns, S. Nole-Wilson, A. Yamaguchi, G. Coupland, B. A. Krizek, D. Wagner, A molecular framework for auxin-mediated initiation of flower primordia. *Dev. Cell* **24**, 271–282 (2013). doi:10.1016/j.devcel.2012.12.017 Medline

96. W. Li, Y. Zhou, X. Liu, P. Yu, J. D. Cohen, E. M. Meyerowitz, LEAFY controls auxin response pathways in floral primordium formation. *Sci. Signal.* **6**, ra23 (2013). Medline

97. O. Nilsson, I. Lee, M. A. Blázquez, D. Weigel, Flowering-time genes modulate the response to LEAFY activity. *Genetics* **150**, 403–410 (1998). doi:10.1093/genetics/150.1.403 Medline

98. A. Serrano-Mislata, K. Goslin, B. Zheng, L. Rae, F. Wellmer, E. Graciet, F. Madueño, Regulatory interplay between *LEAFY*, *APETALA1/CAULIFLOWER* and *TERMINAL FLOWER1*: New insights into an old relationship. *Plant Signal. Behav.* **12**, e1370164 (2017). doi:10.1080/15592324.2017.1370164 Medline

99. C. Liu, J. Zhou, K. Bracha-Drori, S. Yalovsky, T. Ito, H. Yu, Specification of *Arabidopsis* floral meristem identity by repression of flowering time genes. *Development* **134**, 1901–1910 (2007). doi:10.1242/dev.003103 Medline

100. H. Yu, T. Ito, F. Wellmer, E. M. Meyerowitz, Repression of AGAMOUS-LIKE 24 is a crucial step in promoting flower development. *Nat. Genet.* **36**, 157–161 (2004). doi:10.1038/ng1286 Medline

101. Z. Tao, L. Shen, C. Liu, L. Liu, Y. Yan, H. Yu, Genome-wide identification of SOC1 and SVP targets during the floral transition in *Arabidopsis*. *Plant J.* **70**, 549–561 (2012). doi:10.1111/j.1365-313X.2012.04919.x Medline

102. H. Yu, Y. Xu, E. L. Tan, P. P. Kumar, AGAMOUS-LIKE 24, a dosage-dependent mediator of the flowering signals. *Proc. Natl. Acad. Sci. U.S.A.* **99**, 16336–16341 (2002). doi:10.1073/pnas.212624599 Medline

103. J.-W. Wang, B. Czech, D. Weigel, miR156-regulated SPL transcription factors define an endogenous flowering pathway in *Arabidopsis thaliana*. *Cell* **138**, 738–749 (2009). doi:10.1016/j.cell.2009.06.014 Medline

104. S. D. Michaels, E. Himelblau, S. Y. Kim, F. M. Schomburg, R. M. Amasino, Integration of flowering signals in winter-annual *Arabidopsis*. *Plant Physiol.* **137**, 149–156 (2005). doi:10.1104/pp.104.052811 Medline

105. I. Searle, Y. He, F. Turck, C. Vincent, F. Fornara, S. Kröber, R. A. Amasino, G. Coupland, The transcription factor FLC confers a flowering response to vernalization by repressing meristem competence and systemic signaling in *Arabidopsis*. *Genes Dev.* **20**, 898–912 (2006). doi:10.1101/gad.373506 Medline

106. R. Borner, G. Kampmann, J. Chandler, R. Gleissner, E. Wisman, K. Apel, S. Melzer, A MADS domain gene involved in the transition to flowering in *Arabidopsis*. *Plant J.* **24**, 591–599 (2000). doi:10.1046/j.1365-313x.2000.00906.x Medline

107. A. Samach, H. Onouchi, S. E. Gold, G. S. Ditta, Z. Schwarz-Sommer, M. F. Yanofsky, G. Coupland, Distinct roles of CONSTANS target genes in reproductive development of *Arabidopsis*. *Science* **288**, 1613–1616 (2000). doi:10.1126/science.288.5471.1613 Medline

108. S. R. Hepworth, F. Valverde, D. Ravenscroft, A. Mouradov, G. Coupland, Antagonistic regulation of flowering-time gene SOC1 by CONSTANS and FLC via separate promoter motifs. *EMBO J.* **21**, 4327–4337 (2002). doi:10.1093/emboj/cdf432 Medline